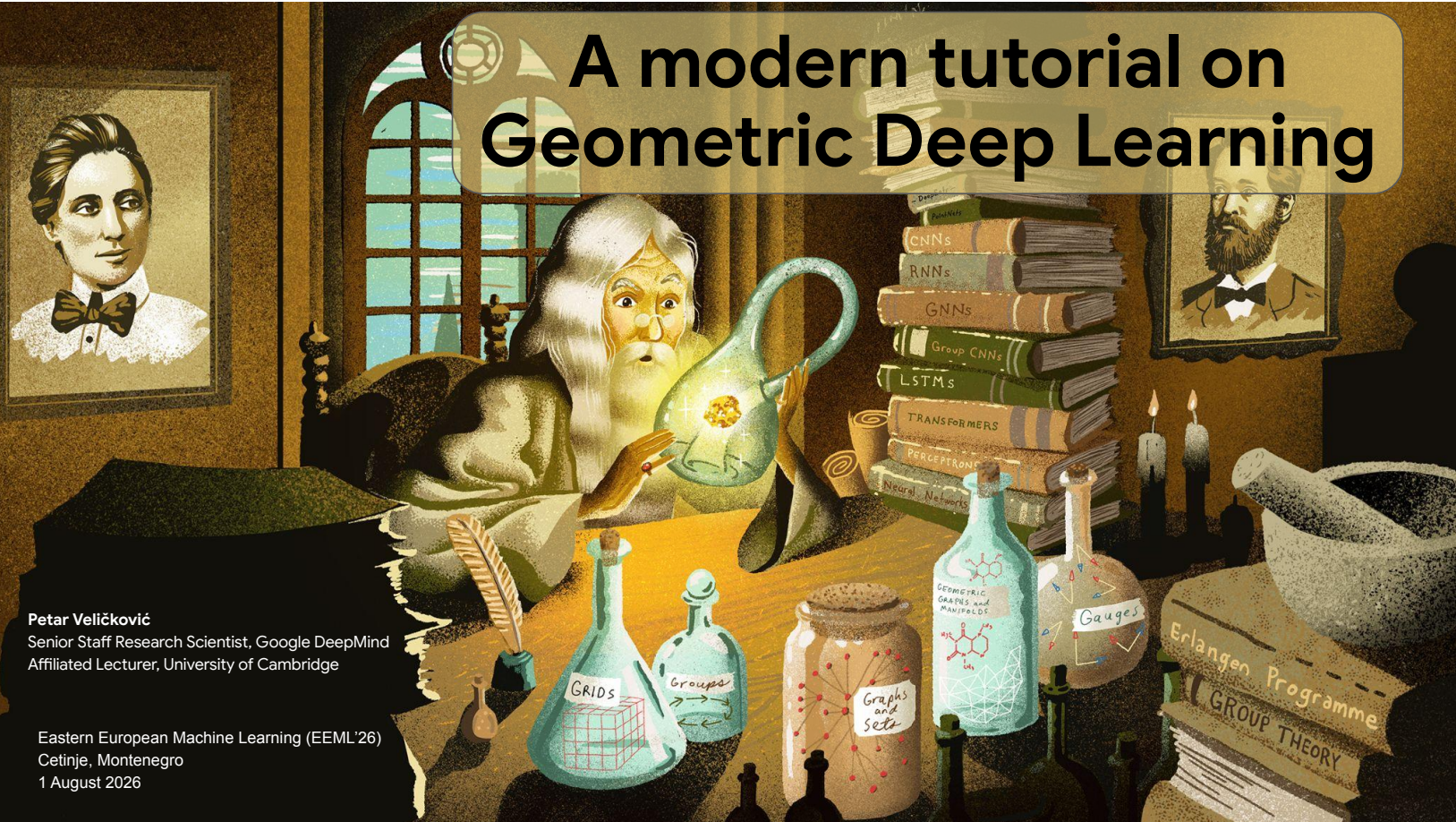
A modern tutorial on Geometric Deep Learning



Petar Veličković
Senior Staff Research Scientist, Google DeepMind
Affiliated Lecturer, University of Cambridge

Eastern European Machine Learning (EEML'26)
Cetinje, Montenegro
1 August 2026

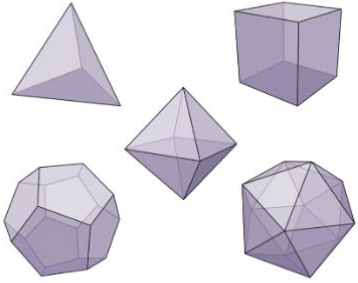
A tale of
symmetry
geometry
groups

“

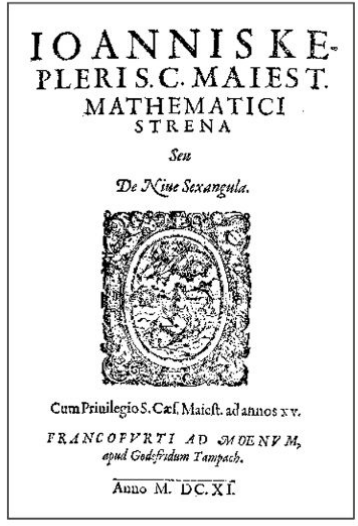
Symmetry, as wide or as narrow as you may define its meaning, is one idea by which man through the ages has tried to comprehend and create **order**, **beauty**, and **perfection**.

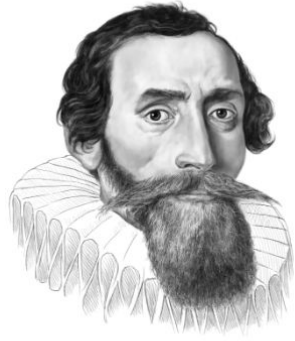
Hermann
Weyl
(1885–1955)

σνμμετρία—*same measure*

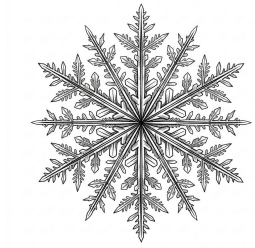


Plato
(ca. 370 BC)

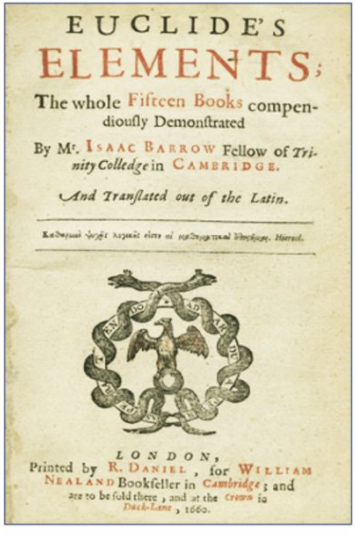




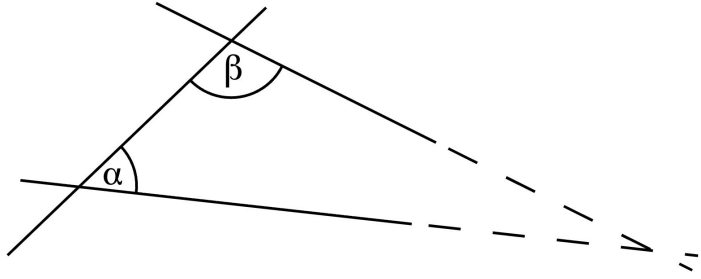
Johannes Kepler
(1571–1630)



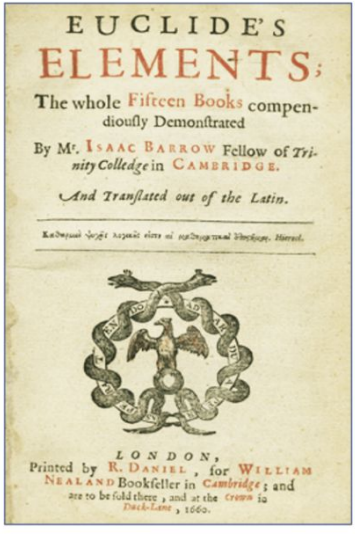
(Euclidean) Geometry


EUCLIDE'S
ELEMENTS;
The whole Fifteen Books compendiously Demonstrated
By Mr. ISAAC BARROW Fellow of Trinity Colledge in CAMBRIDGE.
And Translated out of the Latin.
LONDON,
Printed by R. DANIEL, for WILLIAM NEALAND Bookseller in Cambridge; and are to be sold there, and at the Crown in Duck-Lane, 1660.

Euclid
(ca. 300 BC)



(Euclidean) Geometry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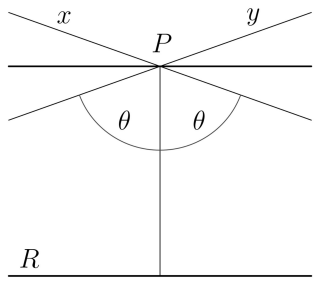
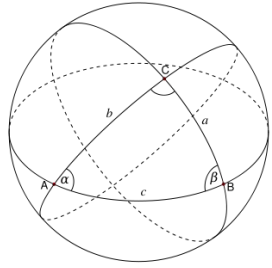
Euclid
(ca. 300 BC)

EUCLIDES
AB OMNI NÆVO VINDICATUS:
SIVE
CONATUS GEOMETRICUS
QUO STABILIUNTUR
Prima ipsa universæ Geometriæ Principia.
AUCTORE
HIERONYMO SACCHERIO
SOCIETATIS JESU
In Ticinensi Universitate Matheseos Professore.
OPUSCULUM
EX.MO SENATUI
MEDIOLANENSI
Ab Auctore Dicatum.
MEDIOLANI, MDCCXXXIII.
Ex Typographia Pauli Antonii Montani. Superiorum permissu.

Omar Khayyam
(1048–1131)

A proper **zoo** of **geometries**





Carl Friedrich Gauss (1777–1855)

János Bolyai (1802–1860)

178

О НАЧАЛАХЪ ГЕОМЕТРІИ (*).

(*Г. Лобачевскаго.*)

Кажется, трудность понятій увеличивается по мѣрѣ ихъ приближенія къ начальнымъ истинамъ въ природѣ; также какъ она возрастаетъ въ другомъ направленіи, къ той границѣ, куда стремится умъ за новыми познаніями. Вотъ почему трудности въ Геометріи должны пренадлежать, вопервыхъ, самому предмету. Далѣе, средства, къ которымъ надобно прибѣгнуть чтобы достигнуть здѣсь послѣдней строгости, едва ли могутъ опытчатъ цѣли и простотѣ сего ученія. Тѣ, которые хотѣли удовлетворить симъ требованіямъ, заключили себя въ такой тѣсной кругъ, что всѣ усилія ихъ не могли быть вознаграждены успѣхомъ. Наконецъ скажемъ и то, что со времени Ньютона и Декарта, вся Математика, слѣвшись Аналитикой, пошла столь быстрыми шагами впередъ, что оставила далеко за собой то ученіе, безъ котораго могла уже об-

(*) Извлечено самимъ Сочинителемъ изъ разсужденія, подъ названіемъ: *Exposition succincte des principes de la Géometrie etc.*, читаннаго имъ въ засѣданіи Отдѣленія физико-математическихъ наукъ, 11 Февраля 1826 года.

Nikolai Lobachevsky (1792–1856)

Bernhard Riemann (1826–1866)

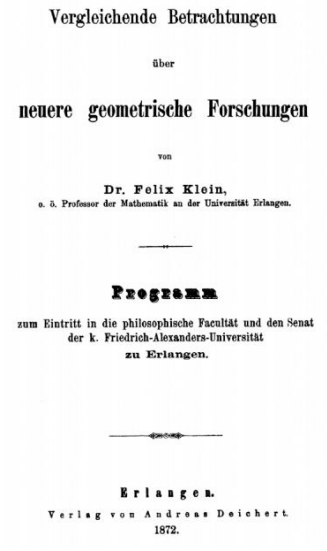
“

I have discovered such wonderful things that I was **amazed**...

out of nothing I have created a **strange new universe**.

János Bolyai (1802–1860)

Organisation came from **group theory**



Vergleichende Betrachtungen

über

neuere geometrische Forschungen

von

Dr. Felix Klein,
o. ö. Professor der Mathematik an der Universität Erlangen.

Programm

zum Eintritt in die philosophische Facultät und den Senat
der k. Friedrich-Alexanders-Universität
zu Erlangen.

Erlangen.
Verlag von Andreas Deichert.
1872.

Felix Klein
(1849–1925)

Erlangen Program

Geometries can be understood by the symmetries they keep

Downstream effects were massive

Élie Cartan (1869–1951)

GENERAL THEORY OF NATURAL EQUIVALENCES

BY

SAMUEL EILENBERG AND SAUNDERS MacLANE

CONTENTS

	Page
Introduction	231
I. Categories and functors	237
1. Definition of categories	237
2. Examples of categories	239
3. Functors in two arguments	241
4. Examples of functors	242
5. Slicing of functors	245
6. Foundations	246
II. Natural equivalence of functors	248
7. Transformations of functors	248
8. Categories of functors	250
9. Composition of functors	250
10. Examples of transformations	251
11. Groups as categories	256
12. Construction of functors by transformations	257
13. Combination of the arguments of functors	258
III. Functors and groups	260
14. Subfunctors	260
15. Quotient functors	262
16. Examples of subfunctors	263
17. The isomorphism theorems	265
18. Direct products of functors	267
19. Characters	270
IV. Partially ordered sets and projective limits	272
20. Quasi-ordered sets	272
21. Direct systems as functors	273
22. Inverse systems as functors	276
23. The categories $\mathfrak{Dir}$ and $\mathfrak{Inv}$	277
24. The lifting principle	280
25. Functors which commute with limits	281
V. Applications to topology	283
26. Complexes	283
27. Homology and cohomology groups	284
28. Duality	287
29. Universal coefficient theorems	288
30. Čech homology groups	290
31. Miscellaneous remarks	292
Appendix. Representations of categories	292

Introduction. The subject matter of this paper is best explained by an example, such as that of the relation between a vector space L and its "dual"

Presented to the Society, September 8, 1942; received by the editors May 15, 1945.

Samuel Eilenberg (1913–1998)

Saunders Mac Lane (1909–2005)

Downstream effects were massive

Invariante Variationsprobleme.

(F. Klein zum fünfzigjährigen Doktorjubiläum.)

Von

Emmy Noether in Göttingen.

Vorgelegt von F. Klein in der Sitzung vom 26. Juli 1918[1]).

Es handelt sich um Variationsprobleme, die eine kontinuierliche Gruppe (im Lieschen Sinne) gestatten; die daraus sich ergebenden Folgerungen für die zugehörigen Differentialgleichungen finden ihren allgemeinsten Ausdruck in den in § 1 formulierten, in den folgenden Paragraphen bewiesenen Sätzen. Über diese aus Variationsproblemen entspringenden Differentialgleichungen lassen sich viel präzisere Aussagen machen als über beliebige, eine Gruppe gestattende Differentialgleichungen, die den Gegenstand der Lieschen Untersuchungen bilden. Das folgende beruht also auf einer Verbindung der Methoden der formalen Variationsrechnung mit denen der Lieschen Gruppentheorie. Für spezielle Gruppen und Variationsprobleme ist diese Verbindung der Methoden nicht neu; ich erwähne Hamel und Herglotz für spezielle endliche, Lorentz und seine Schüler (z. B. Fokker), Weyl und Klein für spezielle unendliche Gruppen[2]). Insbesondere sind die zweite Kleinsche Note und die vorliegenden Ausführungen gegenseitig durch einander beein-

1) Die endgiltige Fassung des Manuskripts wurde erst Ende September eingereicht.

2) Hamel: Math. Ann. Bd. 59 und Zeitschrift f. Math. u. Phys. Bd. 50. Herglotz: Ann. d. Phys. (4) Bd. 36, bes. § 9, S. 511. Fokker, Versl. d. Amsterdamer Akad., 27./1. 1917. Für die weitere Literatur vergl. die zweite Note von Klein: Göttinger Nachrichten 19. Juli 1918.

In naher dem synchronen Arbeit von Kasner (Math. Zeitschrift Bd. 2) handelt es sich um Aufstellung von Invarianten nach ähnlicher Methoden.

Egl. Ges. d. Wiss. Nachrichten. Math.-phys. Klasse, 1918. Heft 3. 17

Emmy Noether (1882–1935)

Hermann Weyl (1885–1955)

Chen-Ning Yang (b. 1922)

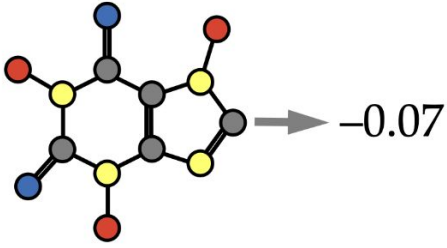
Robert Mills (1927–1999)

A tale of
symmetry
geometry
groups

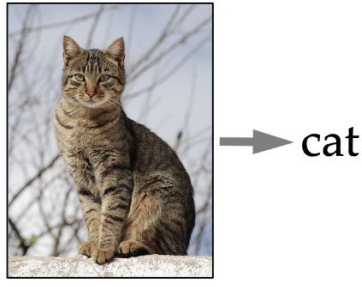
A tale of
symmetry
geometry
groups
& deep learning

Foundations of
Supervised Learning

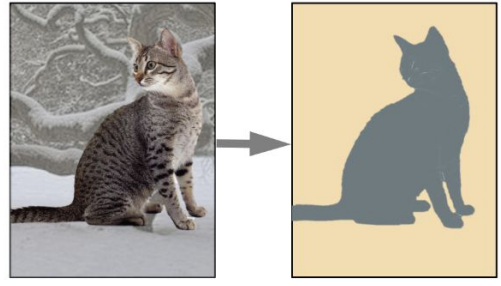
Supervised deep learning tasks



Regression
(solubility log*P*)



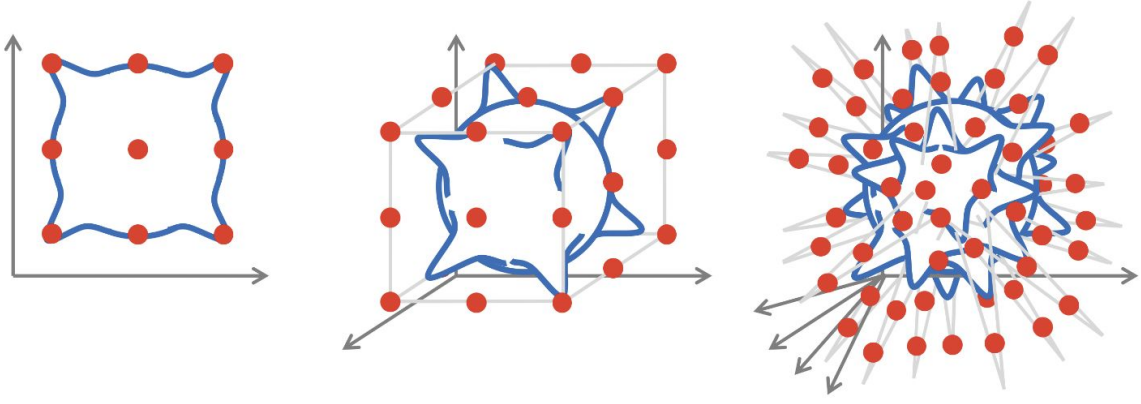
Classification
(binary: cat/dog)



Structured prediction
(image segmentation)

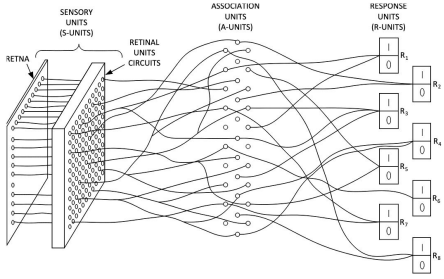
These are real-world, **high dimensional** problems!!!

(hitting the "curse of dimensionality")

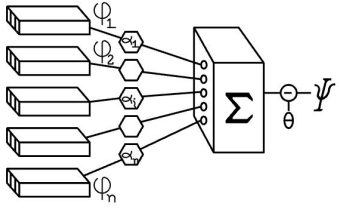


These are real-world,
high dimensional problems!!!
(hitting the "curse of dimensionality")

The space of solutions is **too big**,
and we need **geometry** to restrict it!

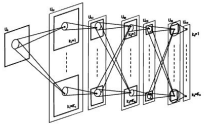


Frank Rosenblatt
(1928–1971)

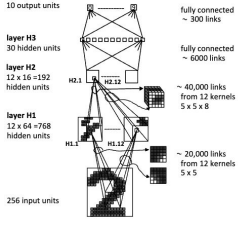


Marvin Minsky
(1927–2016)

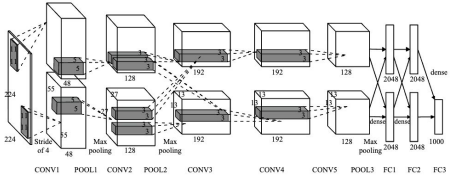
Seymour Papert
(1928–2016)



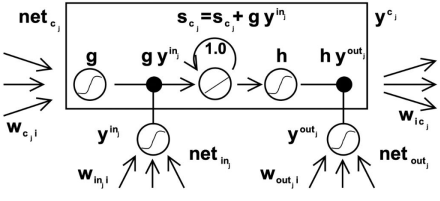
Kunihiko
Fukushima



Yann LeCun

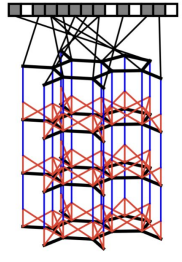


Alex Krizhevsky

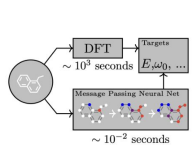


Sepp
Hochreiter

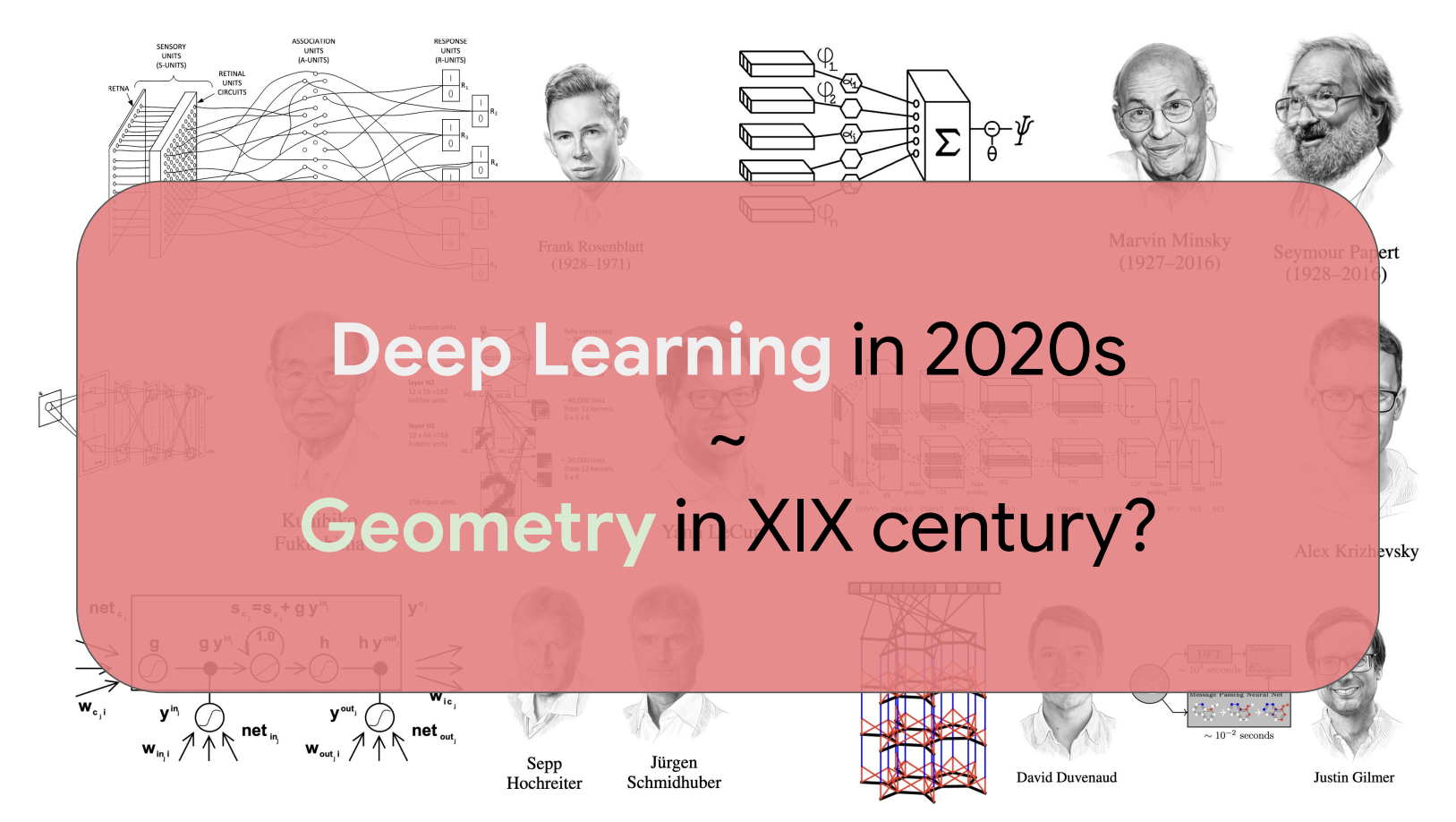
Jürgen
Schmidhuber



David Duvenaud



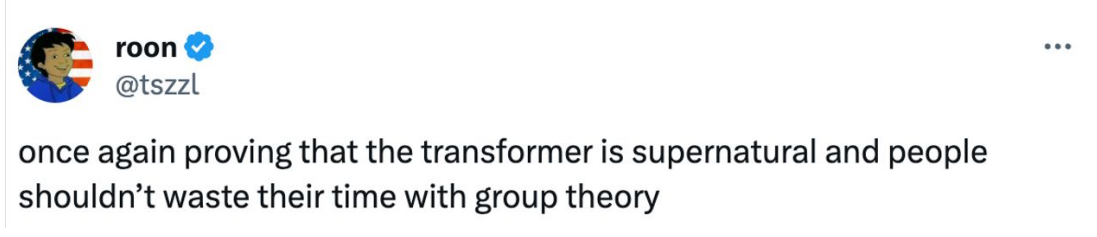
Justin Gilmer


Deep Learning in 2020s
~
Geometry in XIX century?
Frank Rosenblatt
(1928–1971)
Marvin Minsky
(1927–2016)
Seymour Papert
(1928–2016)
Alex Krizhevsky
Sepp
Hochreiter
Jürgen
Schmidhuber
David Duvenaud
Justin Gilmer

Today, we step in Klein's shoes!

Use **group theory** to make (**geometric**) sense of **deep learning architectures** through the **symmetries** they keep

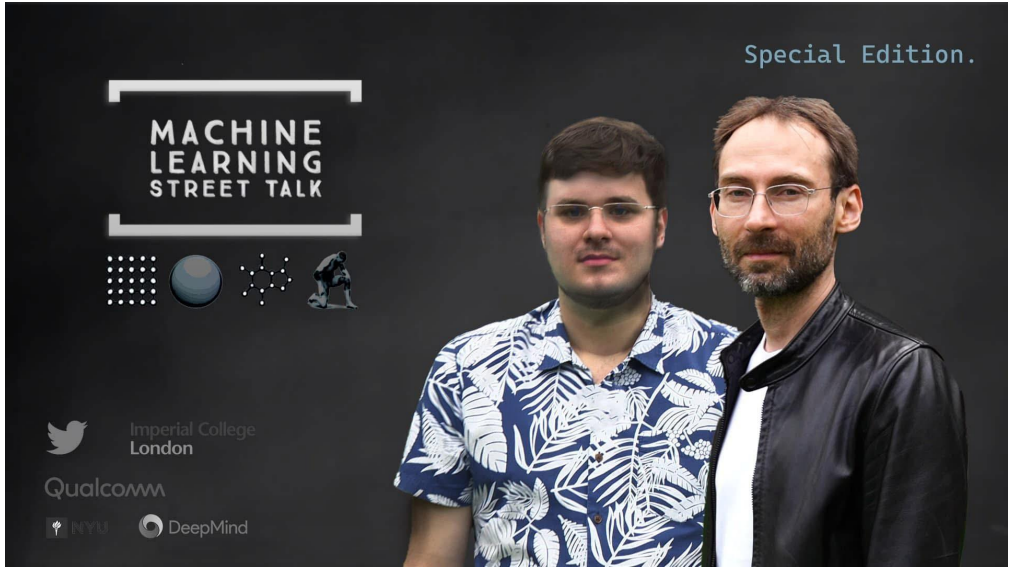
...or should we?


roon
@tszzl
once again proving that the transformer is supernatural and people shouldn't waste their time with group theory

Vee is based today
@Wyrvyrn
staggering number of great engineers and mathematicians tragically nerdsniped by gdl 😔

The current status quo

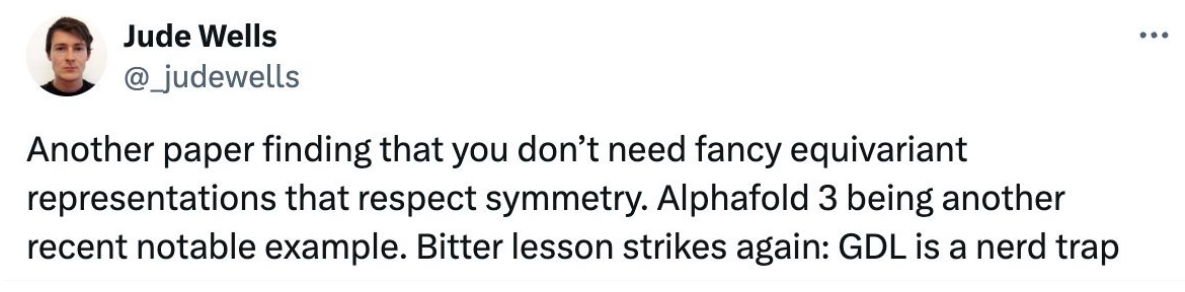
Geometric Deep Learning had its wildest surge in 2021–23, when most saw it as a way to build new 3D architectures, largely for **molecular** data



The current status quo

Geometric Deep Learning had its wildest surge in 2021–23, when most saw it as a way to build new 3D architectures, largely for **molecular** data

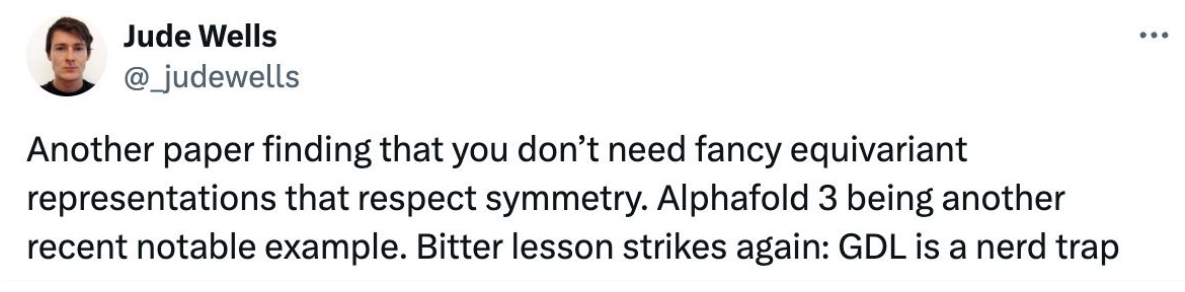
...that hype is now all but over :)



The current status quo

Geometric Deep Learning had its wildest surge in 2021–23, when most saw it as a way to build new 3D architectures, largely for **molecular** data

...that hype is now all but over :)

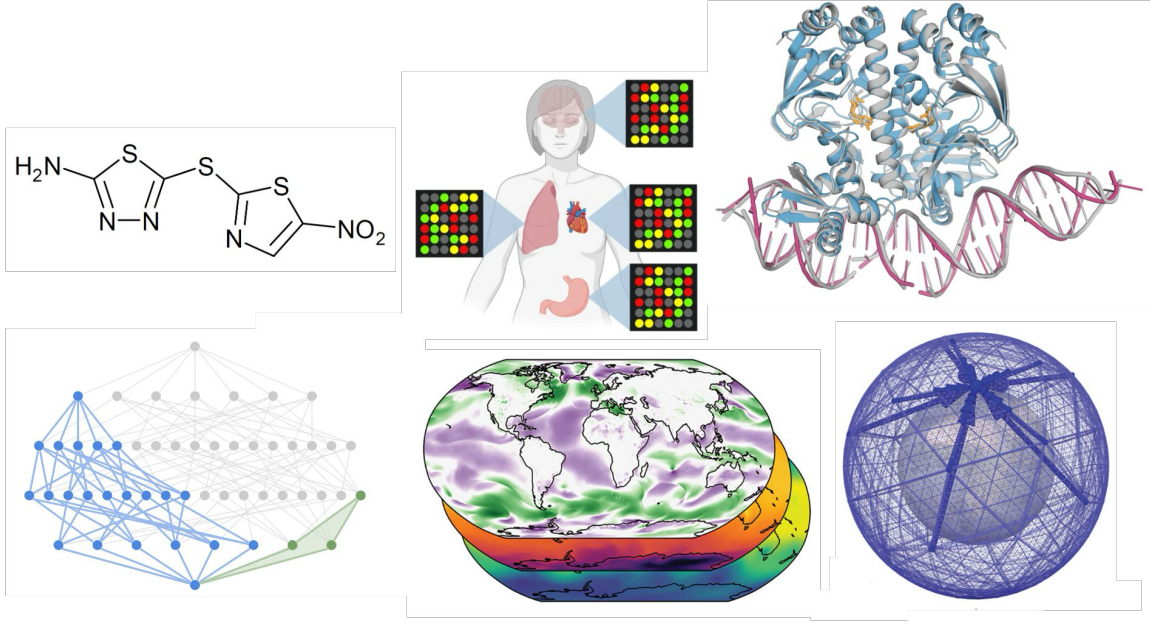


Once the hype elements are stripped away, what remains?

Why learn GDL in 2026?

Why learn GDL in 2026?

1. Data from **nature** is geometric—good to be mindful of it



Why learn GDL in 2026?

1. Data from **nature** is geometric—good to be mindful of it
2. Keep honest about **constraints** we assume, offering perspective

Contemporary LLMs:

Cut input into **tokens**

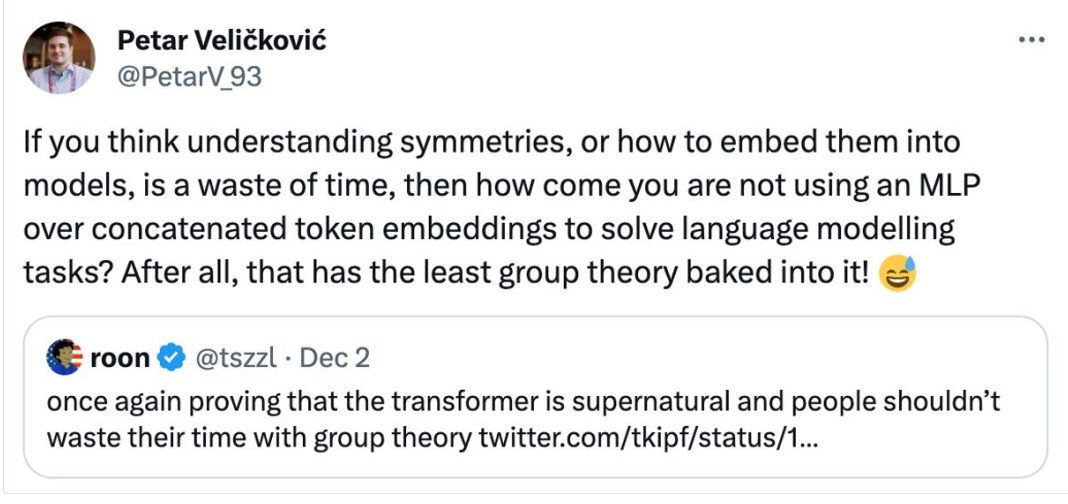
Modify each token embedding **separately** w/ **shared** function

Only allow token interactions through **shared attention**

WHY?

Why learn GDL in 2026?

1. Data from **nature** is geometric—good to be mindful of it
2. Keep honest about **constraints** we assume, offering perspective



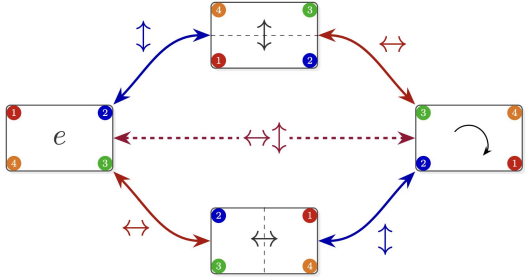
Why learn GDL in 2026?

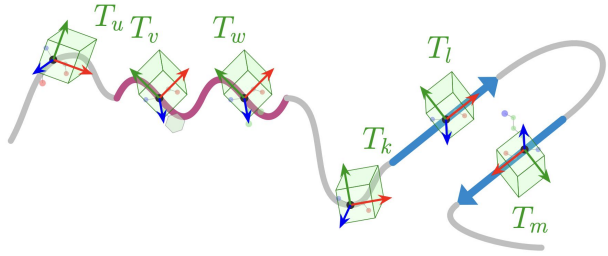
1. Data from **nature** is geometric—good to be mindful of it
2. Keep honest about **constraints** we assume, offering perspective
3. Sometimes, can derive **exactly** how to build certain neural nets

$$\mathbf{F}(\mathbf{X}) = \begin{bmatrix} \theta_0 & \theta_1 & & & \theta_{-1} \\ \theta_{-1} & \theta_0 & \theta_1 & & \\ & \ddots & \ddots & \ddots & \\ & & \theta_{-1} & \theta_0 & \theta_1 \\ \theta_1 & & & \theta_{-1} & \theta_0 \end{bmatrix} \begin{bmatrix} — & \mathbf{x}_0 & — \\ — & \mathbf{x}_1 & — \\ & \vdots & \\ — & \mathbf{x}_{n-2} & — \\ — & \mathbf{x}_{n-1} & — \end{bmatrix}$$

Why learn GDL in 2026?

1. Data from **nature** is geometric—good to be mindful of it
2. Keep honest about **constraints** we assume, offering perspective
3. Sometimes, can derive **exactly** how to build certain neural nets
4. Following GDL principles can lead to cool & useful **architectures** :)





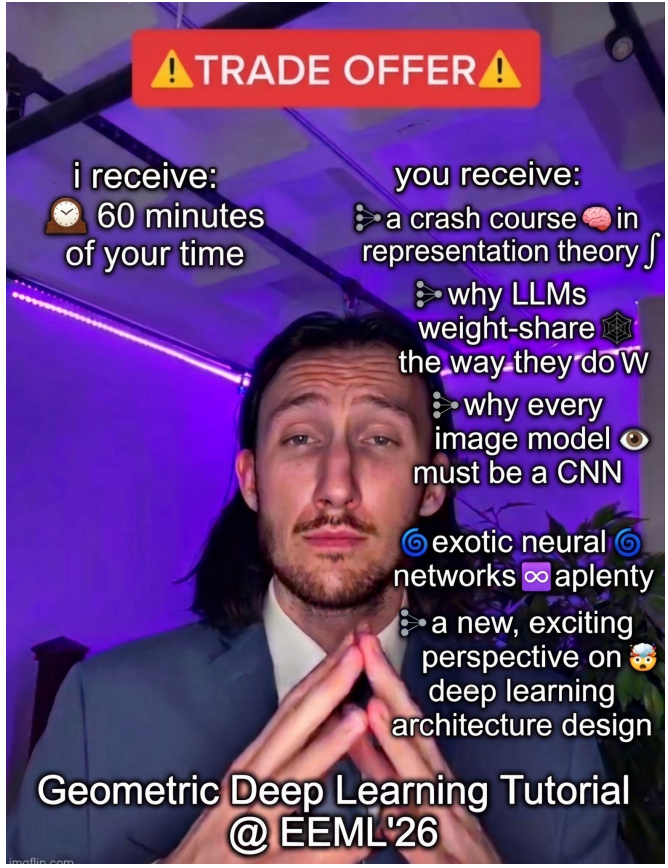
What will we cover today?

1. Data from **nature** is geometric—good to be mindful of it
   → Learn **symmetry**, **representations**, **equivariance** essentials

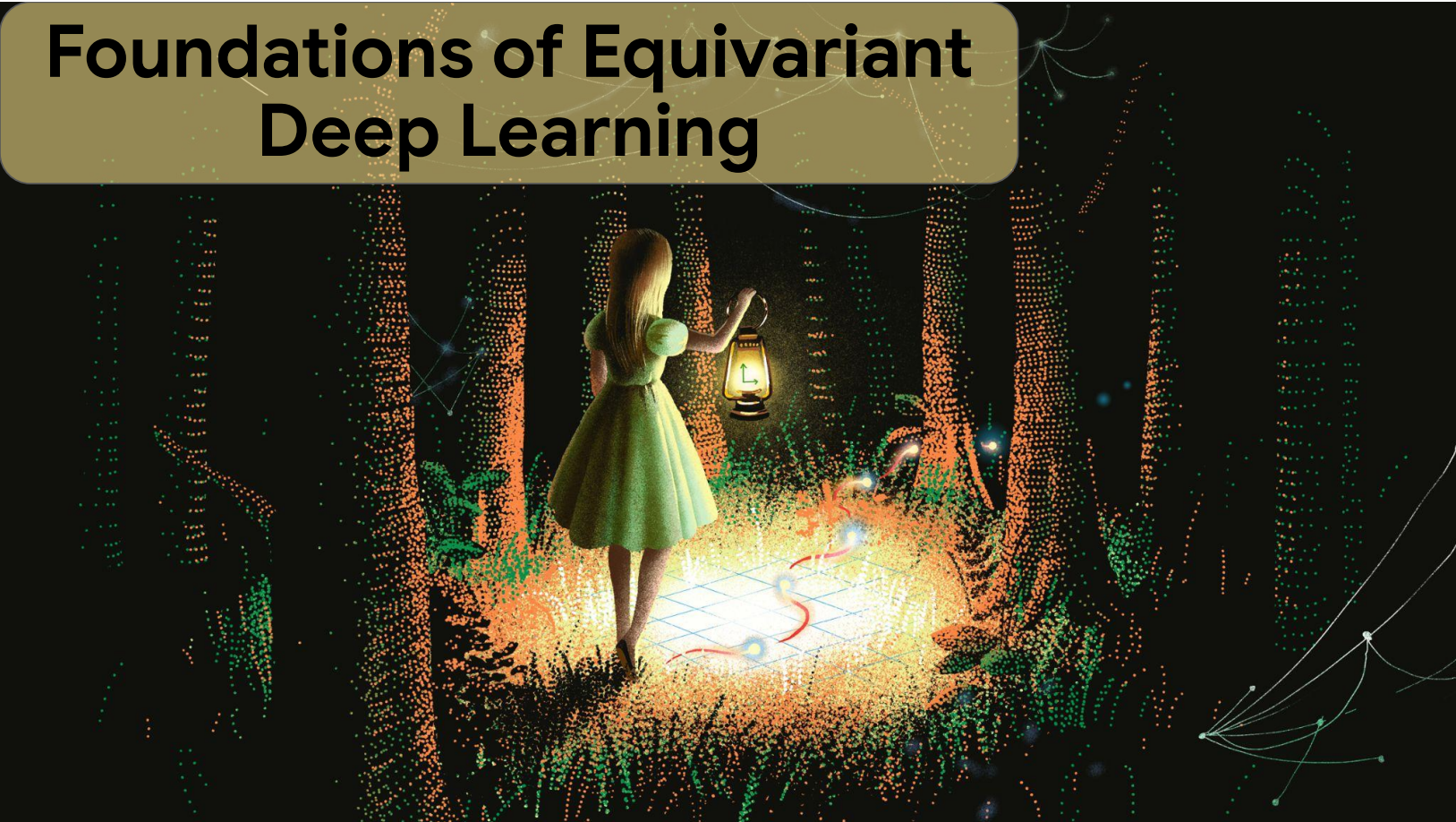
2. Keep honest about **constraints** we assume, offering perspective
   → Explain why **Transformers** have the constraints they do

3. Sometimes, can derive **exactly** how to build certain neural nets
   → Show how **CNNs** must have the form they do

4. Following GDL principles can lead to cool & useful **architectures** :)
   → Teach how to build NNs over **spheres**, **meshes** & *special guests*!

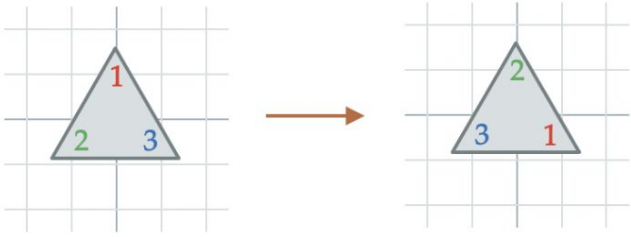

⚠️TRADE OFFER⚠️
i receive:
🕰️ 60 minutes of your time
you receive:
🔌a crash course🧠in representation theory ∫
🔌why LLMs weight-share 🕸️ the way they do W
🔌why every image model 👁️ must be a CNN
🌀exotic neural 🌀 networks ♾️aplenty
🔌a new, exciting perspective on 🤯 deep learning architecture design
Geometric Deep Learning Tutorial @ EEML'26
imgflip.com

Foundations of Equivariant Deep Learning

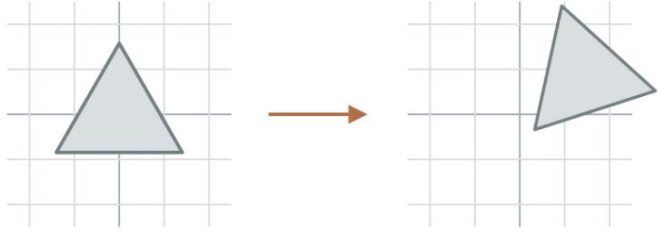


A *symmetry transformation* (or automorphism) of an object is an invertible and structure-preserving map from the object to itself.

120° Rotation

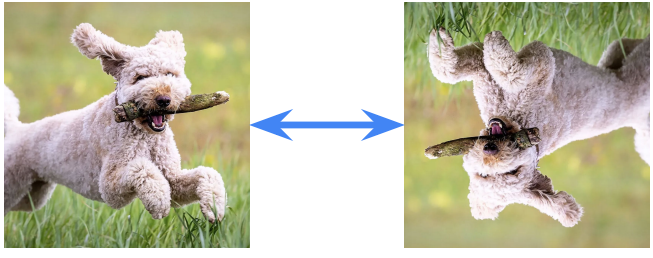


General isometry

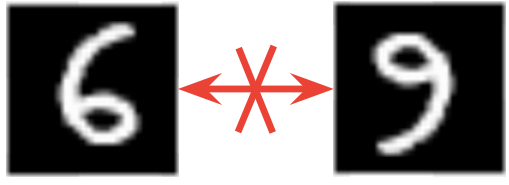


A *symmetry transformation* (or automorphism) of an object is an invertible and structure-preserving map from the object to itself.

A symmetry in cats vs. dogs



Not a symmetry in digit recognition



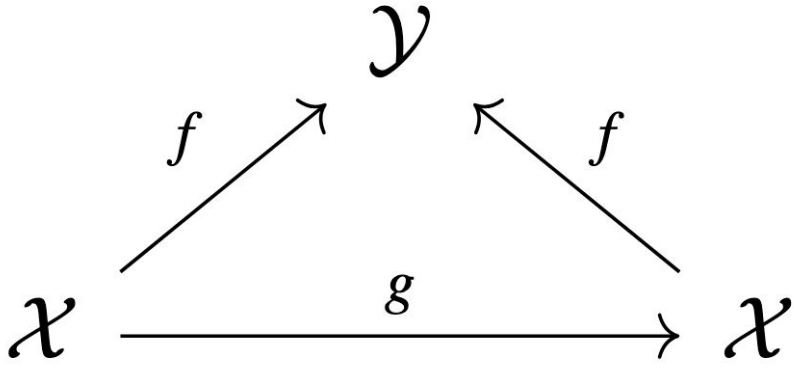
A *symmetry transformation* (or automorphism) of an object is an invertible and structure-preserving map from the object to itself.

The **same** object might have very ***different*** **symmetries** depending on specifics of the problem! (‘6’ vs. ‘9’)

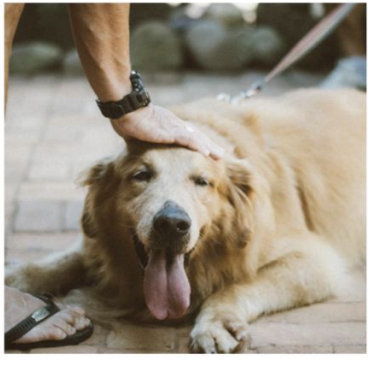
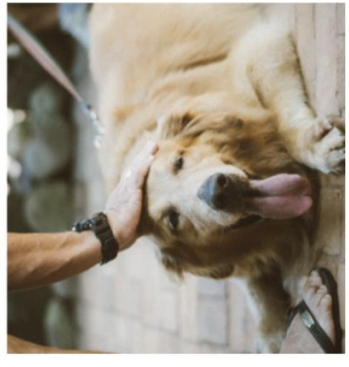
$$f : \mathcal{X} \to \mathcal{Y}$$

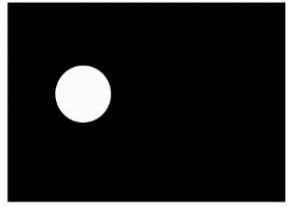
$f : \mathcal{X} \to \mathcal{Y}$ Invariant to symmetry $g : \mathcal{X} \to \mathcal{X}$

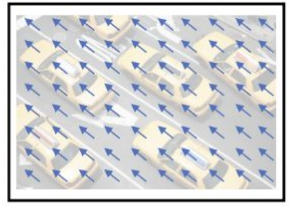
$f : \mathcal{X} \to \mathcal{Y}$ **Invariant** to **symmetry** $g : \mathcal{X} \to \mathcal{X}$



$f : \mathcal{X} \to \mathcal{Y}$ Invariant to symmetry $g : \mathcal{X} \to \mathcal{X}$

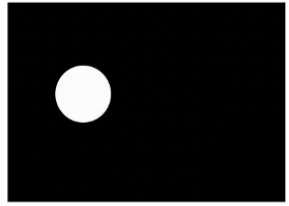
f() = f() = “dog”

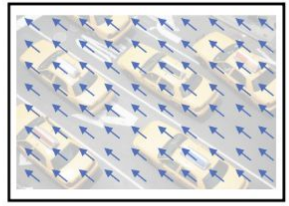




Invariant

Equivariant





$$f : \mathcal{X} \to \mathcal{Y}$$

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Equivariant
to symmetry

$$\rho_{\mathcal{X}}(g) : \mathcal{X} \to \mathcal{X}$$
$$\rho_{\mathcal{Y}}(g) : \mathcal{Y} \to \mathcal{Y}$$

$f($ $) =$

$$f : \mathcal{X} \to \mathcal{Y}$$

Equivariant to **symmetry**

$$\rho_{\mathcal{X}}(g) : \mathcal{X} \to \mathcal{X}$$
$$\rho_{\mathcal{Y}}(g) : \mathcal{Y} \to \mathcal{Y}$$

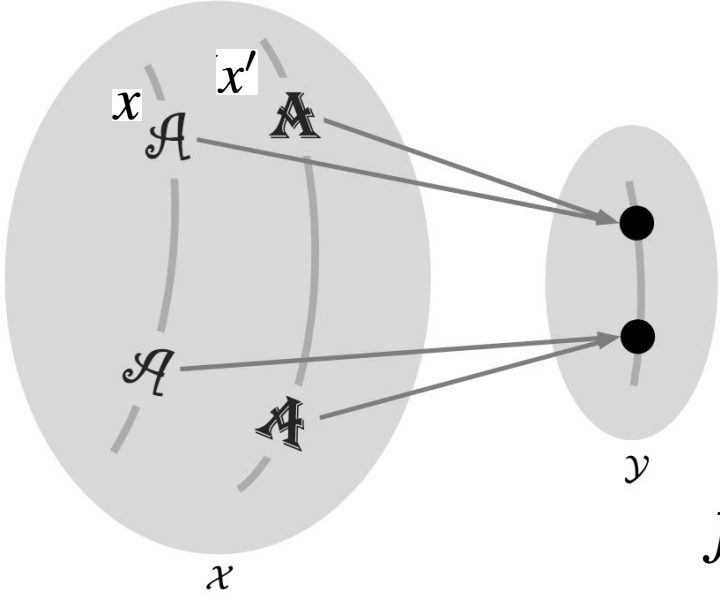
$$\begin{array}{ccc} \mathcal{Y} & \xrightarrow{\rho_{\mathcal{Y}}(g)} & \mathcal{Y} \\ {\scriptstyle f^*}\uparrow & & \uparrow{\scriptstyle f^*} \\ \mathcal{X} & \xrightarrow{\rho_{\mathcal{X}}(g)} & \mathcal{X} \end{array}$$

Invariance is just a special case of ρ_Y!

$$f : \mathcal{X} \to \mathcal{Y}$$

Equivariant
to symmetry

$$\rho_{\mathcal{X}}(g) : \mathcal{X} \to \mathcal{X}$$
$$\rho_{\mathcal{Y}}(g) : \mathcal{Y} \to \mathcal{Y}$$



$$f(x) = f(x')$$

+Equivariance of *f*,
Implies:

$$f(\rho_{\mathcal{X}}(g)x) = f(\rho_{\mathcal{X}}(g)x')$$

To organise **symmetries**, we need **groups**

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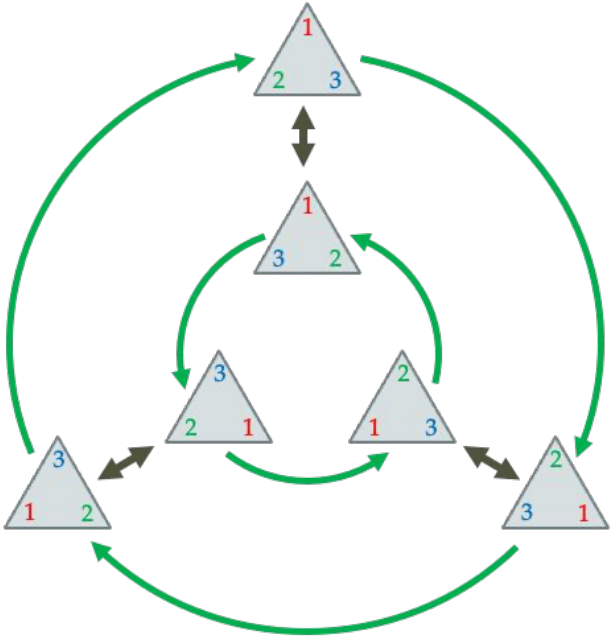
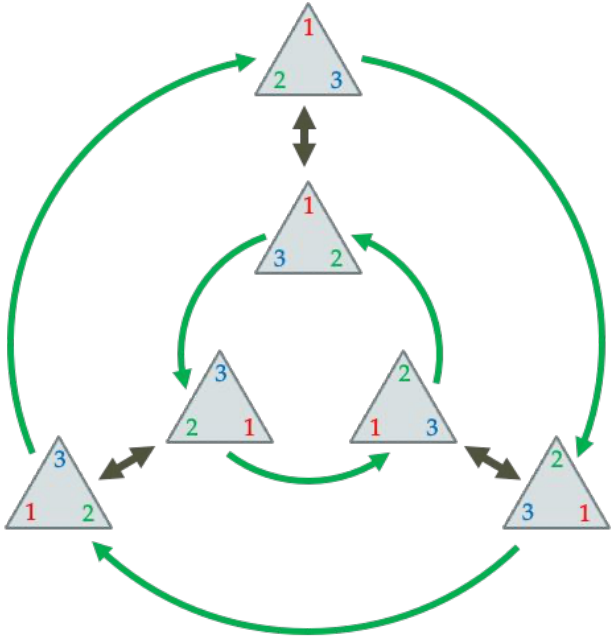
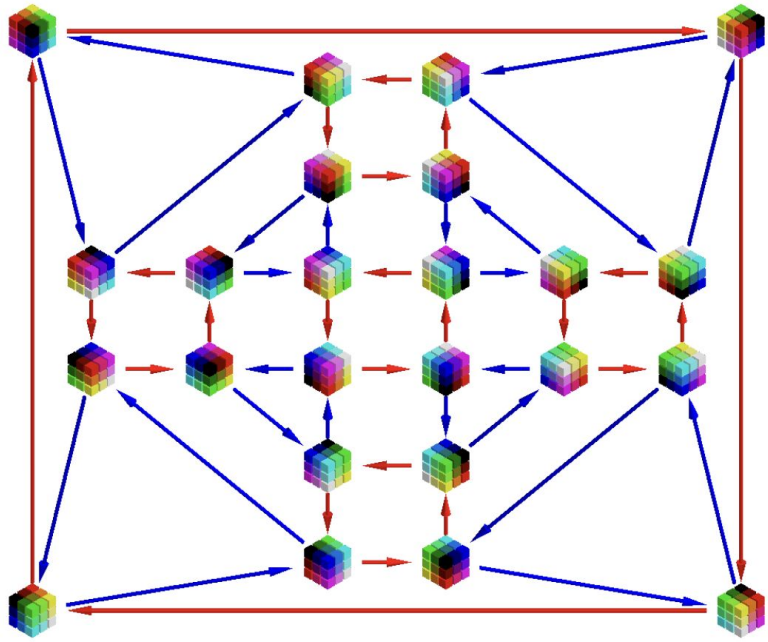


To organise **symmetries**, we need **groups**





To organise **symmetries**, we need **groups**

A *group* is a set $\mathcal{G}$ along with a binary operation $\circ : \mathcal{G} \times \mathcal{G} \to \mathcal{G}$ called *composition* or the *group operation* (for brevity, denoted by juxtaposition $g \circ h = gh$) satisfying the following axioms:

Associativity: $(gh)k = g(hk)$ for all $g, h, k \in \mathcal{G}$.

Identity: there exists a unique $e \in \mathcal{G}$ satisfying $eg = ge = g$ for all $g \in \mathcal{G}$.

Inverse: For each $g \in \mathcal{G}$ there is a unique inverse $g^{-1} \in \mathcal{G}$ such that $gg^{-1} = g^{-1}g = e$.

Turning groups from **abstract objects** into actions

Let $\mathcal{G}$ be a group and let $\mathcal{X}$ be a set. A group action of $\mathcal{G}$ on $\mathcal{X}$ is a mapping $\alpha : \mathcal{G} \times \mathcal{X} \to \mathcal{X}$ satisfying:

$$\alpha(e, x) = x \qquad \text{(Identity)}$$

$$\alpha(gh, x) = \alpha(g, \alpha(h, x)), \qquad \text{(Composition)}$$

for all $g, h \in \mathcal{G}$, $x \in \mathcal{X}$, and where $e \in \mathcal{G}$ is the identity.

Turning group actions into matrices (representations)

Let $\mathcal{G}$ be a group, let $\mathcal{X}$ be a vector space, and denote by $[\mathcal{X}, \mathcal{X}]$ the set of linear mappings from $\mathcal{X}$ to $\mathcal{X}$. A linear representation (or **Vect**-representation) of $\mathcal{G}$ on $\mathcal{X}$ is a mapping $\rho : \mathcal{G} \to [\mathcal{X}, \mathcal{X}]$ that assigns to each $g \in \mathcal{G}$ a linear mapping $\rho(g) : \mathcal{X} \to \mathcal{X}$ satisfying:

$$\rho(e) = \mathrm{id}_{\mathcal{X}} \qquad \text{(Identity)}$$

$$\rho(g) \circ \rho(h) = \rho(gh), \qquad \text{(Composition)}$$

for all $g, h \in \mathcal{G}$, and where $e \in \mathcal{G}$ is the identity.

Some standard **representations**

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SO(2):
2D rotations

$$\rho(\theta) = \begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix}$$

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$$\rho(\theta) = \begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix}$$

S_3:
three-object permutations

$$\rho(\pi_{12}) = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Armed with **representations** we can define **equivariance**

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Let $f : \mathcal{X} \to \mathcal{Y}$ be a mapping of sets (or vector spaces), $\mathcal{G}$ a group, and let $\rho_1 : \mathcal{G} \to [\mathcal{X}, \mathcal{X}]$ and $\rho_2 : \mathcal{G} \to [\mathcal{Y}, \mathcal{Y}]$ be two (linear) representations of $\mathcal{G}$ acting on the domain $\mathcal{X}$ and codomain $\mathcal{Y}$ of f. Then f is *$\mathcal{G}$-equivariant* if for all $g \in \mathcal{G}$:

$$f \circ \rho_1(g) = \rho_2(g) \circ f. \tag{3.18}$$

An equivariant (linear) map can also be called a mapping or morphism of representations, written $f : (\mathcal{X}, \rho_1) \to (\mathcal{Y}, \rho_2)$.

Foundations of Geometric
Deep Learning

Where does the **data** live? **Domain**, Ω

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NxN **images**: $\Omega_N = \left\{ \left(\frac{i}{N}, \frac{j}{N} \right) \right\}_{1 \leq i,j \leq N}$

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Sets: $\Omega = \{1, \ldots, N\}$

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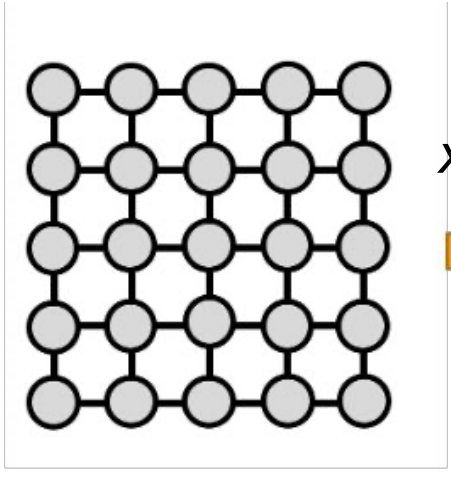
NxN **images**: $\Omega_N = \left\{ \left(\frac{i}{N}, \frac{j}{N} \right) \right\}_{1 \leq i,j \leq N}$

Sets: $\Omega = \{1, \ldots, N\}$

Graphs: $\Omega = G = (\mathcal{V}, \mathcal{E})$

Where does the **data** live? **Signal**, $x : \Omega \rightarrow C$



$\Omega = \mathbb{Z}_n \times \mathbb{Z}_n$

$\mathcal{C} = \mathbb{R}^3$

Actions on **domains** (Ω) imply actions on signals (*x*)!

$$\mathcal{G} \times \Omega \to \Omega \qquad \mathcal{G} \times \mathcal{X} \to \mathcal{X}$$

$$(g, u) \mapsto g \cdot u \qquad (g, x) \mapsto \rho(g)x$$

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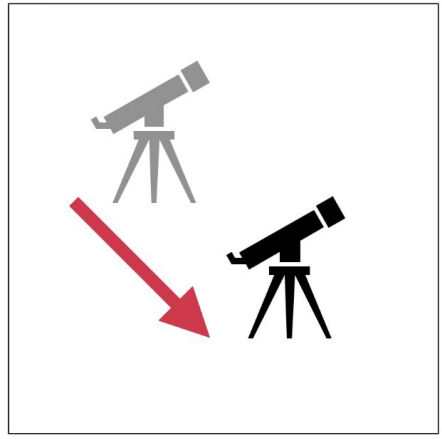
$$(\rho(g)x)(u) := x(g^{-1} \cdot u)$$

Actions on **domains** (Ω) imply actions on signals (x)!

x

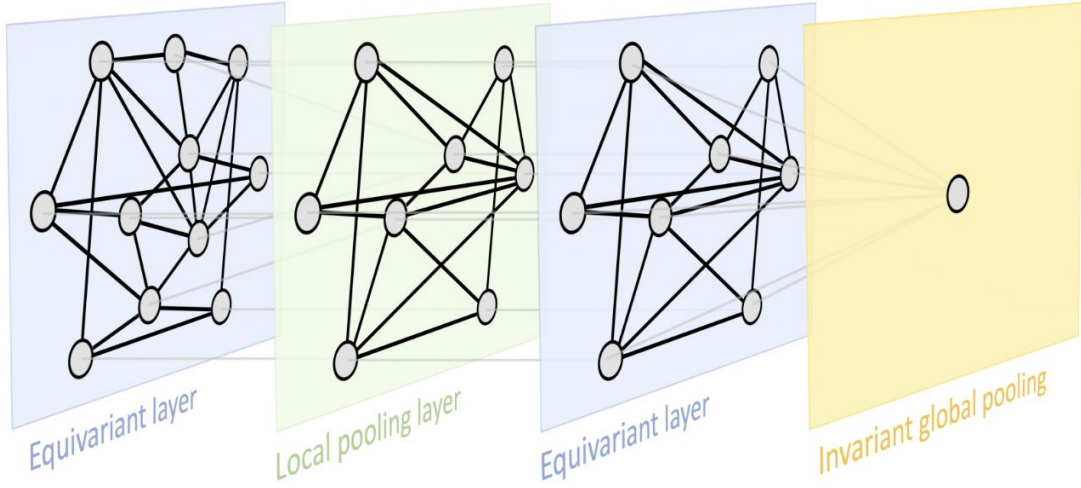


$\rho(g)x$

$$(\rho(g)x)(u) := x(g^{-1} \cdot u)$$

Building blocks of GDL

Since invariants throw away structure, focus on linear **equivariants** (+ **nonlinearities**). Delay invariance until the end:



We can derive all the fan favourites, and more :)

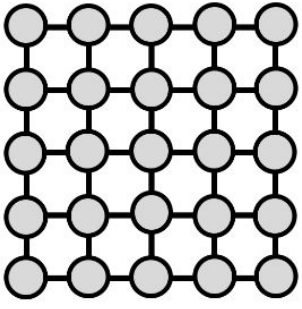
Architecture	**Domain Ω**	**Symmetry group $\mathcal{G}$**
CNN	Grid	Translation
Spherical CNN	Sphere / SO(3)	Rotation SO(3)
Mesh CNN	Manifold	Isometry Iso(Ω) / Gauge symmetry SO(2)
GNN	Graph	Permutation Σ_n
Deep Sets	Set	Permutation Σ_n
Transformer	Complete Graph	Permutation Σ_n
E(3) *GNNs*	Geometric Graph	Permutation Σ_n $\times$ Euclidean E(3)
LSTM	1D Grid	Time warping

We can derive all the fan favourites, and more :)

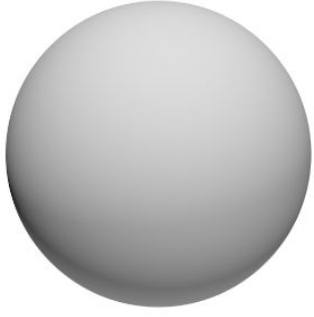
To understand the **geometry** of our data, we need to specify a **domain**, **signal**, **symmetry group** and **representation**

We can use this to make the architecture **explicitly** equivariant, **approximately** equivariant (via data augmentation or loss), or **evaluate** its equivariance preservation (a strong diagnostic)!

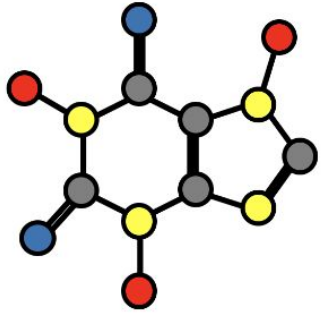
In what remains, we focus on the **“4 Gs”**



Grids



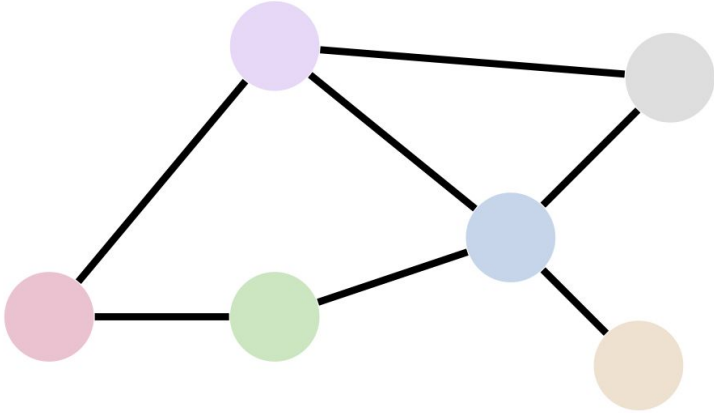
Groups

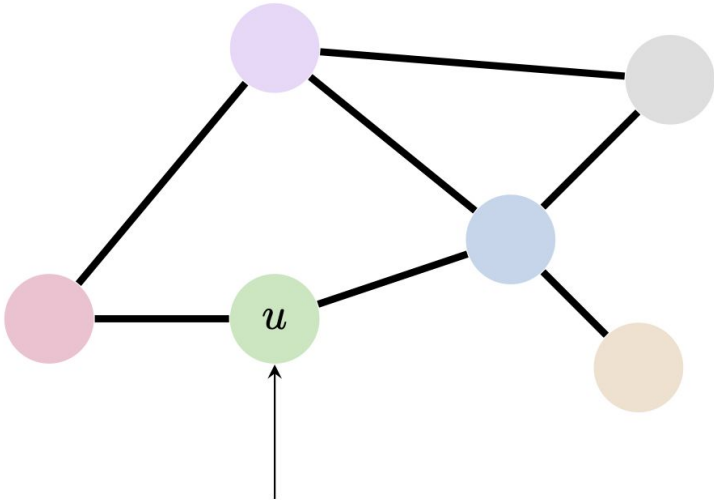


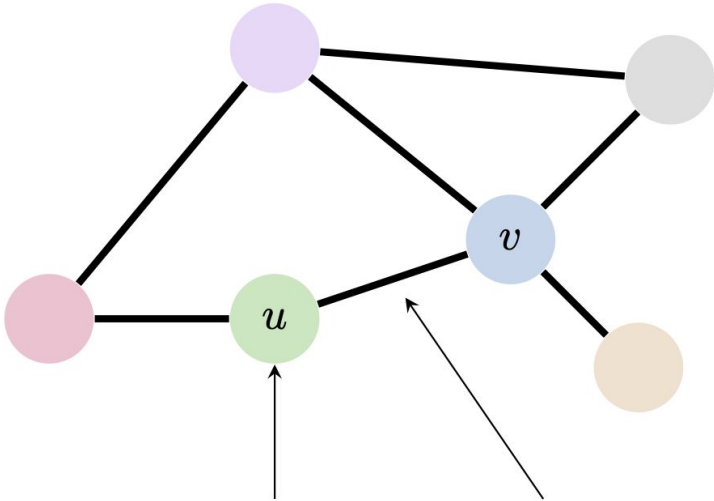
Graphs

Geodesics & Gauges

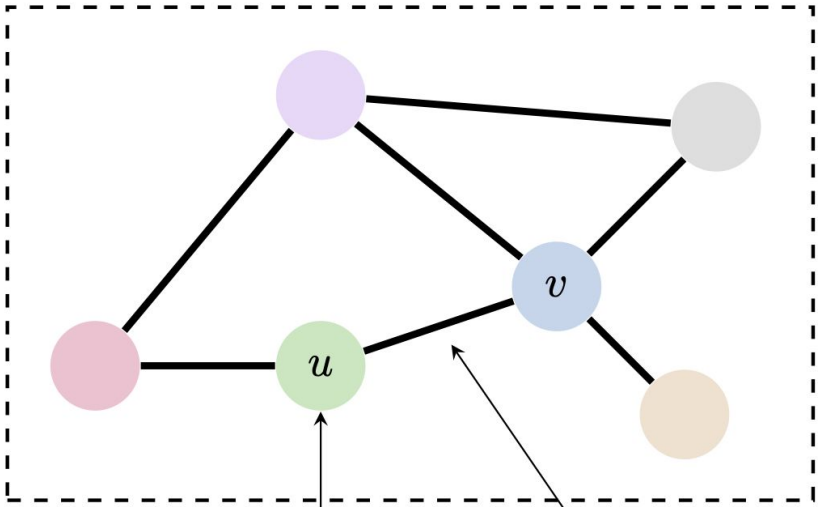
Graphs




u
This
is a node.
$u \in \mathcal{V}$


v
u
This
is a node.
$u \in \mathcal{V}$
This
is an edge.
$(u, v) \in \mathcal{E}$

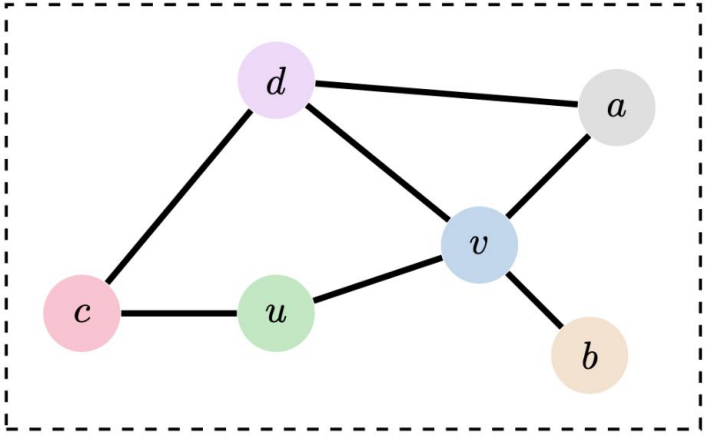
This is a graph. $\mathcal{G} = (\mathcal{V}, \mathcal{E})$



This is a node. $u \in \mathcal{V}$

This is an edge. $(u, v) \in \mathcal{E}$

This is a graph. $\mathcal{G} = (\mathcal{V}, \mathcal{E})$



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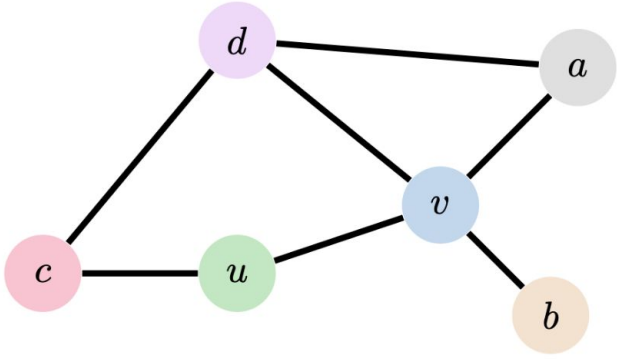


This is its tensorial representation.

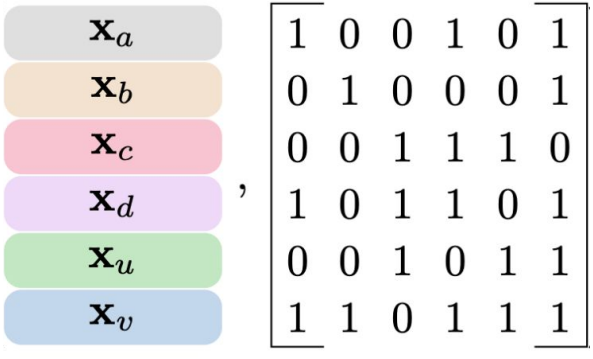
$$\mathbf{X} = \begin{bmatrix} \mathbf{x}_a \\ \mathbf{x}_b \\ \mathbf{x}_c \\ \mathbf{x}_d \\ \mathbf{x}_u \\ \mathbf{x}_v \end{bmatrix} \qquad \mathbf{A} = \begin{bmatrix} 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 1 & 1 & 0 \\ 1 & 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 & 1 & 1 \end{bmatrix}$$

$\mathbf{X} \in \mathbb{R}^{|\mathcal{V}| \times k}$ $\qquad$ $\mathbf{A} \in \mathbb{R}^{|\mathcal{V}| \times |\mathcal{V}|}$

Assuming no features in edges, ***wlog***

The choice of ordering *[a, b, c, d, u, v]* in the tensor is **arbitrary**

Changing it does not change the graph structure!



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$$f\left(\begin{matrix}\mathbf{x}_a\\ \mathbf{x}_b\\ \mathbf{x}_c\\ \mathbf{x}_d\\ \mathbf{x}_u\\ \mathbf{x}_v\end{matrix}, \begin{bmatrix}1 & 0 & 0 & 1 & 0 & 1\\ 0 & 1 & 0 & 0 & 0 & 1\\ 0 & 0 & 1 & 1 & 1 & 0\\ 1 & 0 & 1 & 1 & 0 & 1\\ 0 & 0 & 1 & 0 & 1 & 1\\ 1 & 1 & 0 & 1 & 1 & 1\end{bmatrix}\right) = \mathbf{y}_{\mathcal{G}}$$

The choice of ordering *[a, b, c, d, u, v]* in the tensor is **arbitrary**

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$$f\left(\begin{bmatrix}\mathbf{x}_a\\ \mathbf{x}_b\\ \mathbf{x}_c\\ \mathbf{x}_d\\ \mathbf{x}_u\\ \mathbf{x}_v\end{bmatrix}, \begin{bmatrix}1&0&0&1&0&1\\ 0&1&0&0&0&1\\ 0&0&1&1&1&0\\ 1&0&1&1&0&1\\ 0&0&1&0&1&1\\ 1&1&0&1&1&1\end{bmatrix}\right) = \mathbf{y}_{\mathcal{G}} = f\left(\begin{bmatrix}\mathbf{x}_u\\ \mathbf{x}_v\\ \mathbf{x}_b\\ \mathbf{x}_a\\ \mathbf{x}_c\\ \mathbf{x}_d\end{bmatrix}, \begin{bmatrix}1&1&0&0&1&0\\ 1&1&1&1&0&1\\ 0&1&1&0&0&0\\ 0&1&0&1&0&1\\ 1&0&0&0&1&1\\ 0&1&0&1&1&1\end{bmatrix}\right)$$

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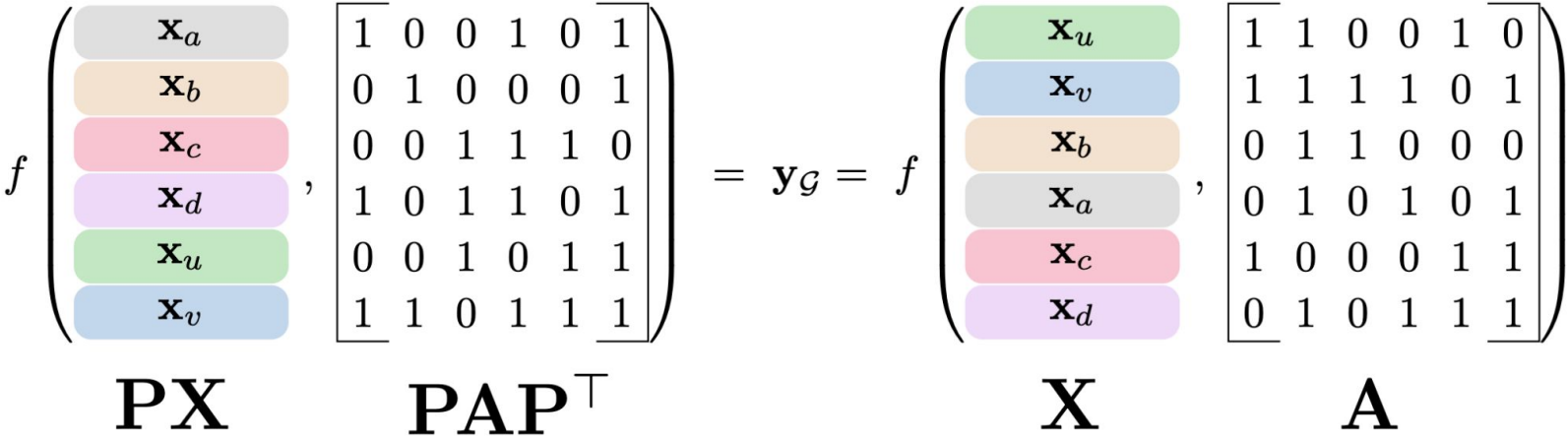
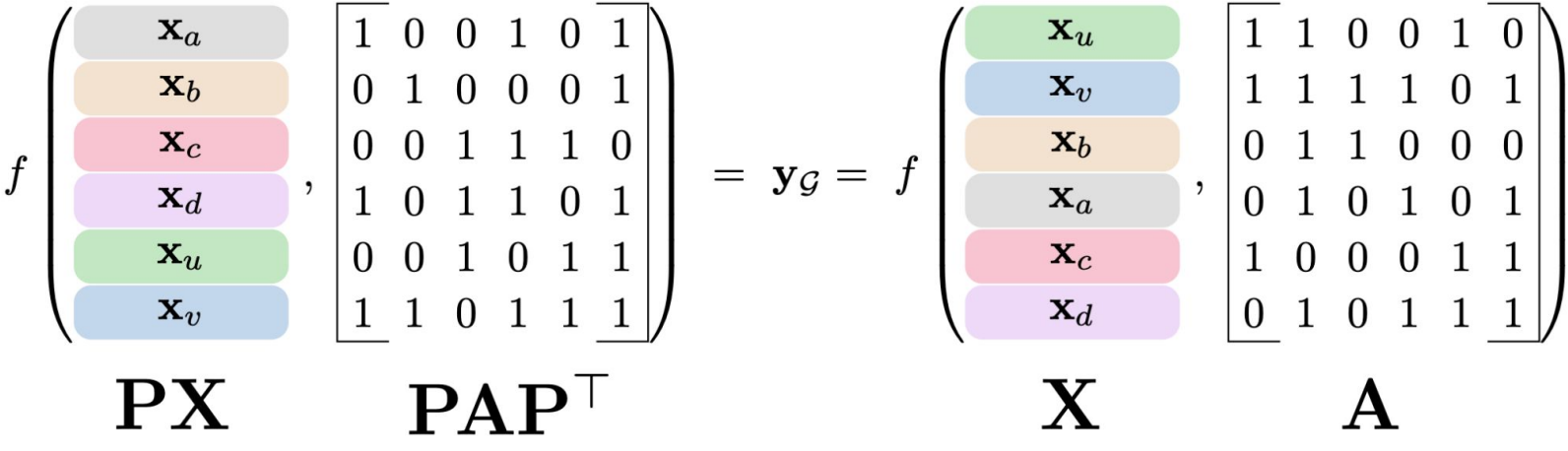


Permutation invariance

Representations are *permutation matrices* **P** = ρ(π), for all permutations π ∈ S_n

From invariance to **equivariance**

Invariance forces **single** output for the graph – what if we want output **per node**?



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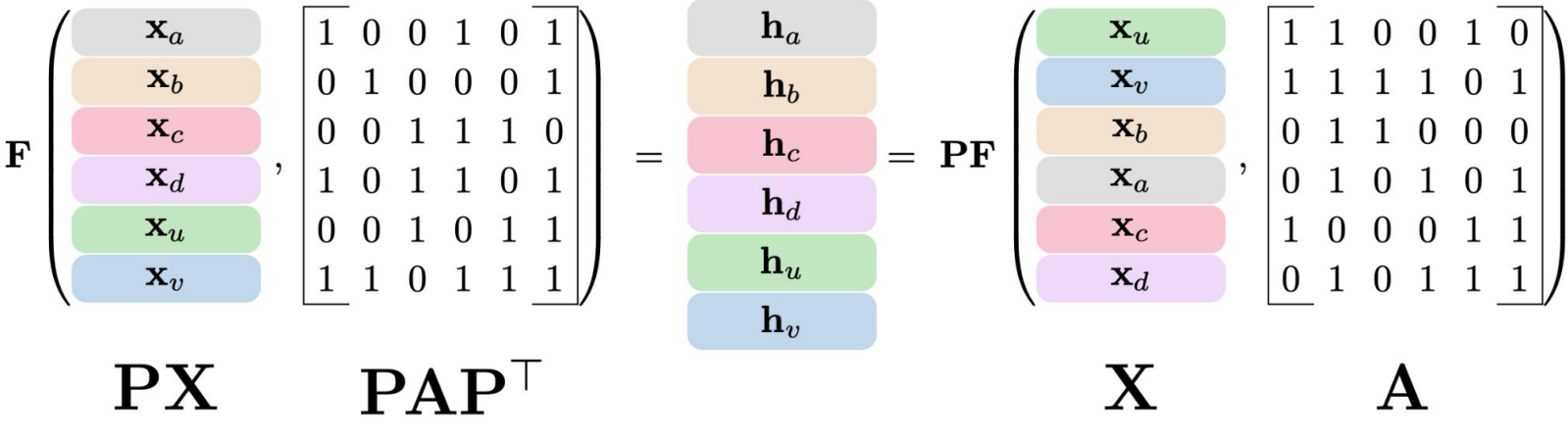

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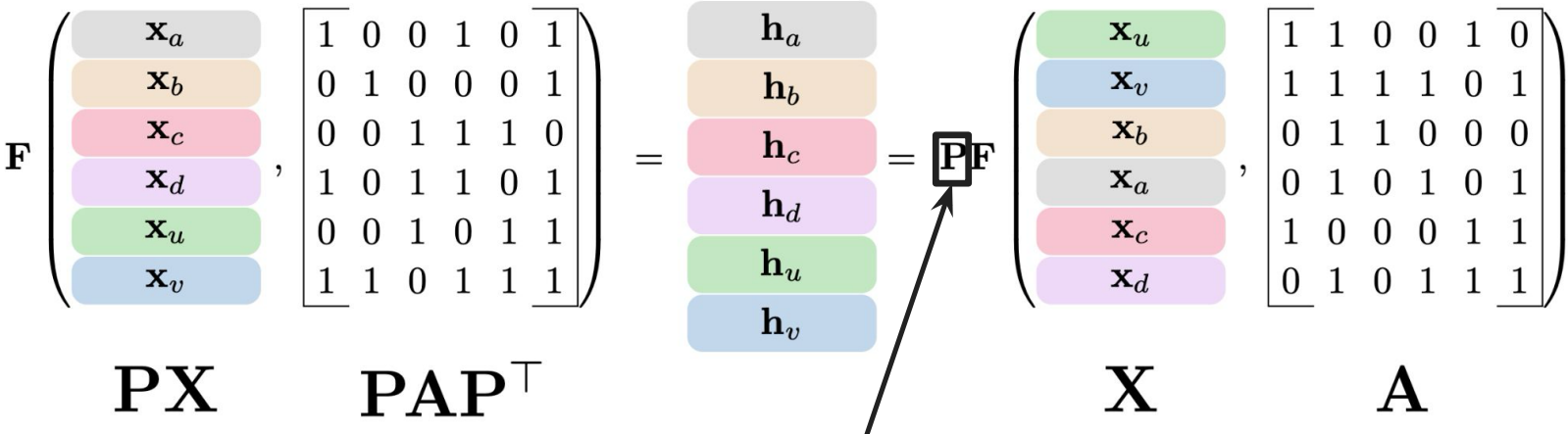
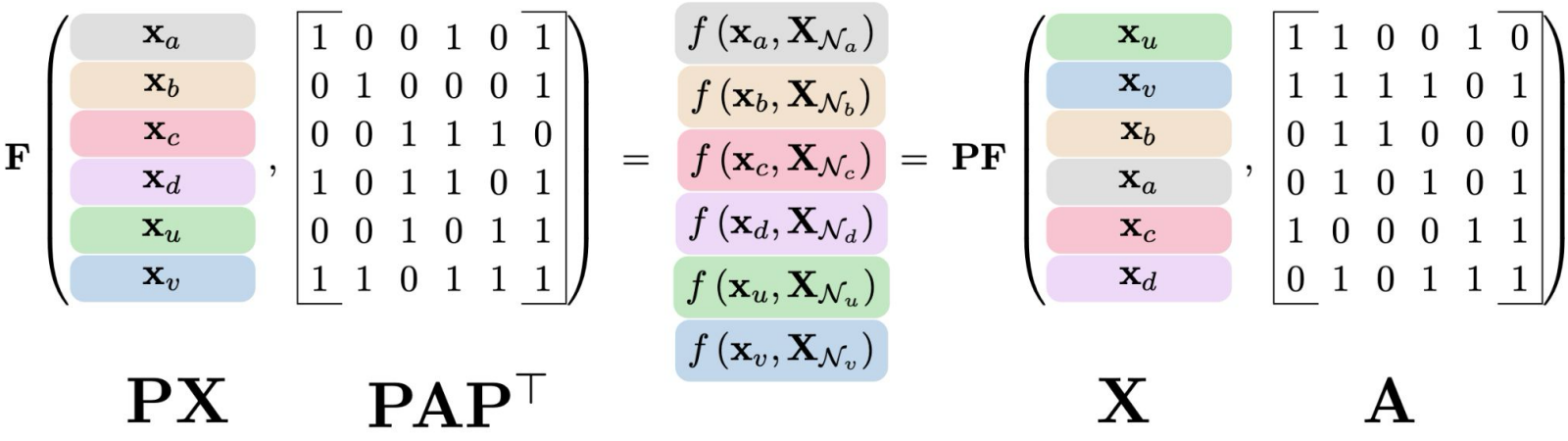


Now the output must change, **equivariantly** – according to the permutation!

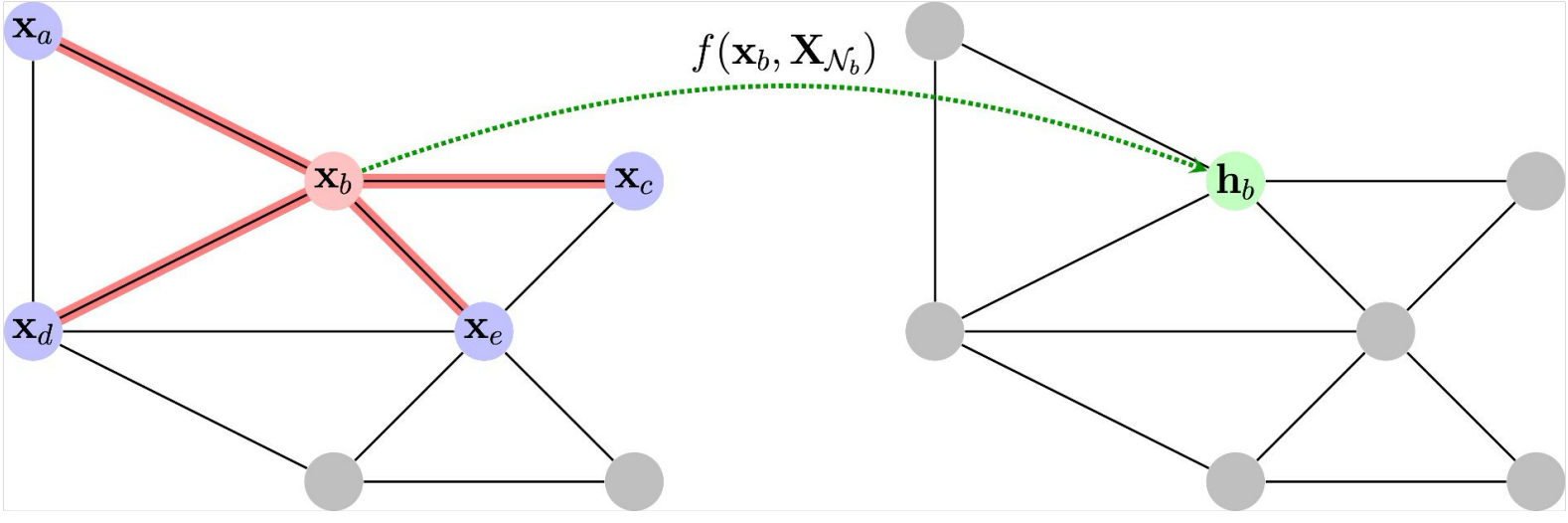
Equivariant models through **locality**

One *(only?!)* way to satisfy this is by making the layer *local* in each node



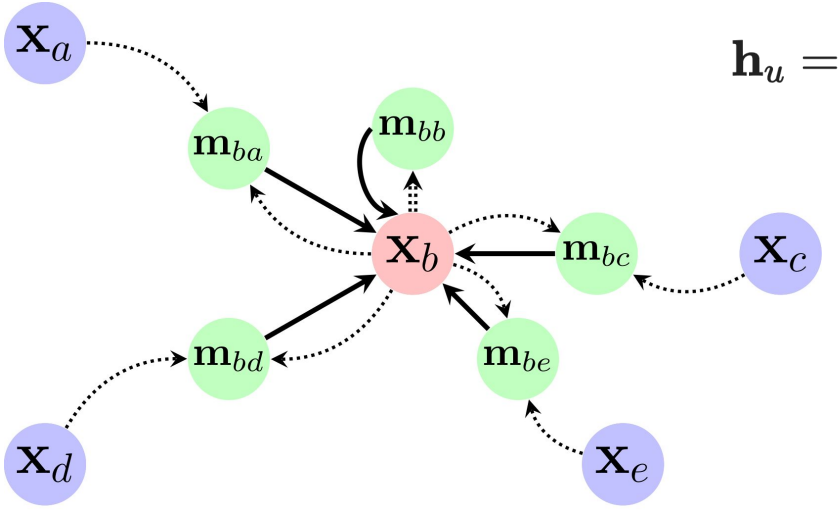
f is any *permutation **invariant*** NN, and $\mathbf{X}_{Nu}$ are all the neighbour features of $\mathbf{x}_u$

Recipe for GNNs, visualised



$$\mathbf{X}_{\mathcal{N}_b} = \{\{\mathbf{x}_a, \mathbf{x}_b, \mathbf{x}_c, \mathbf{x}_d, \mathbf{x}_e\}\}$$

Zooming in: Message-passing neural networks



Message-passing

$$\mathbf{h}_u = \phi\left(\mathbf{x}_u, \bigoplus_{v \in \mathcal{N}_u} \psi\left(\mathbf{x}_u, \mathbf{x}_v\right)\right)$$

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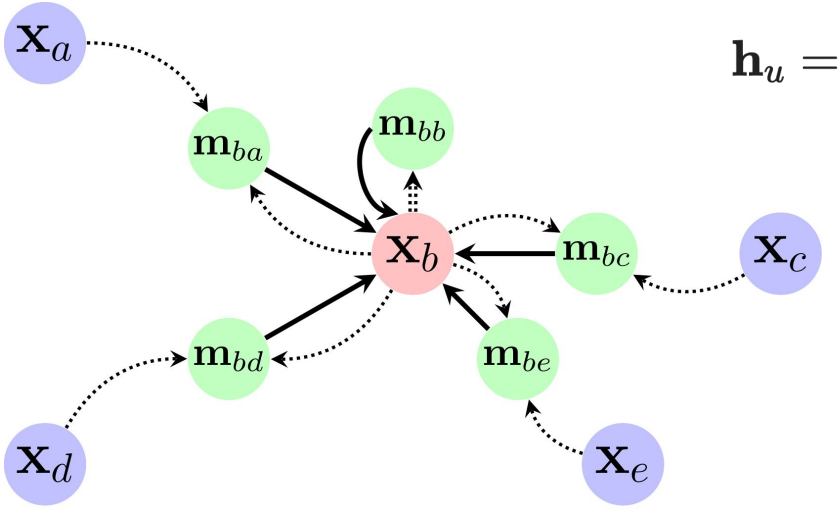

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Message function

$$\psi : \mathbb{R}^k \times \mathbb{R}^k \to \mathbb{R}^m$$

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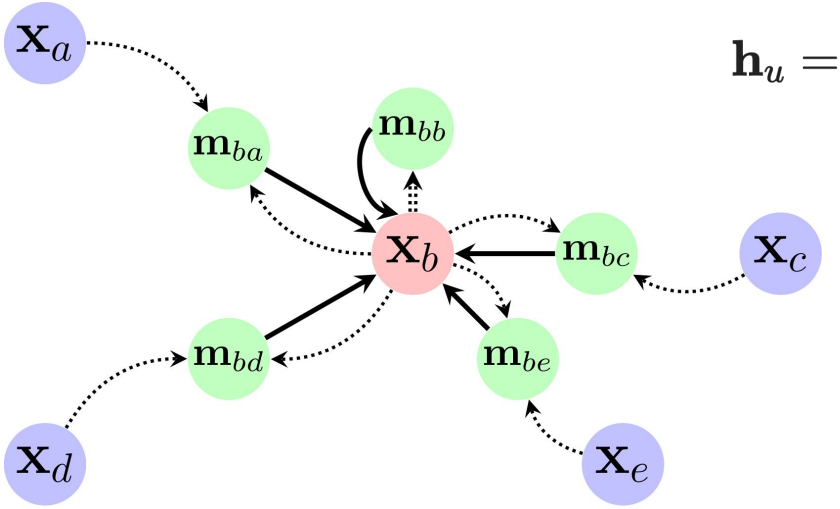

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Aggregation function

$$\bigoplus : \text{bag}(\mathbb{R}^m) \to \mathbb{R}^m$$

The aggregation function, ⊕, needs to be **permutation invariant** (e.g. sum, max, average)

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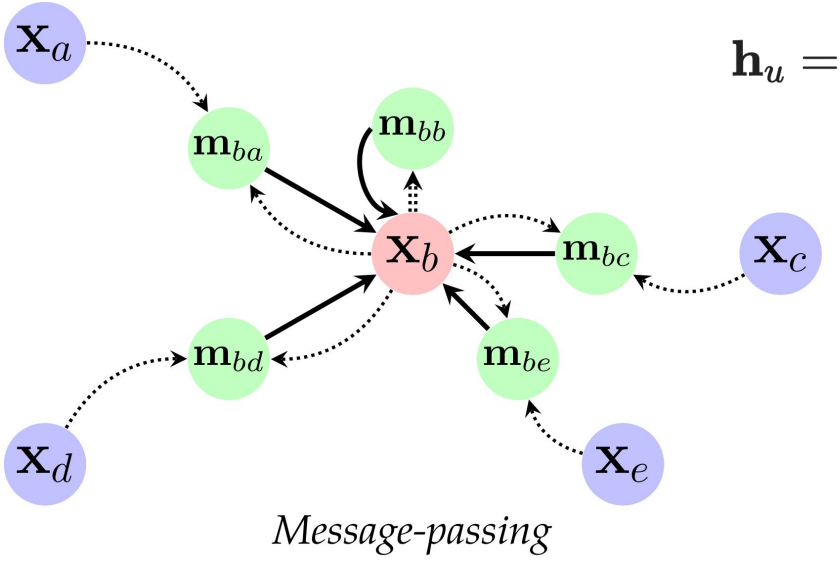

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Update function

$$\phi : \mathbb{R}^k \times \mathbb{R}^m \to \mathbb{R}^l$$

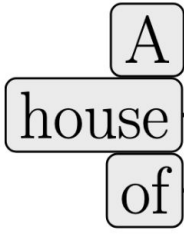
Message and update functions are the only **parametric** components of the message-passing GNN

What’s in an **LLM**?

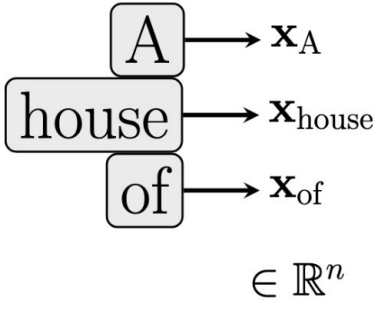
Many people think of modern LLMs as ***sequence*** models...

...but they are **not**!!!!!!!

LLMs as GNNs

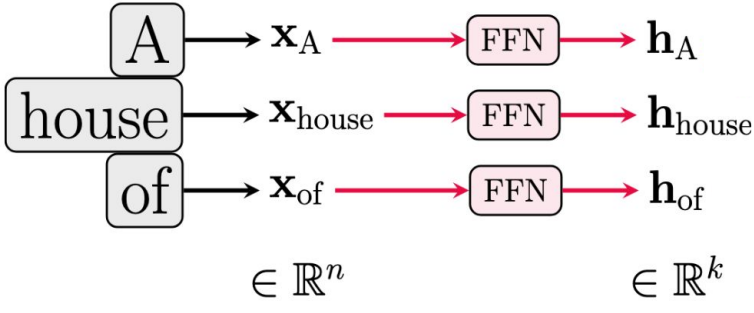


LLMs as GNNs


A
house
of
$\mathbf{x}_{\text{A}}$
$\mathbf{x}_{\text{house}}$
$\mathbf{x}_{\text{of}}$
$\in \mathbb{R}^n$

LLMs as GNNs

$$\mathrm{FFN}(\mathbf{x}) = \mathbf{W}_2\sigma(\mathbf{W}_1\mathbf{x} + \mathbf{b}_1) + \mathbf{b}_2$$



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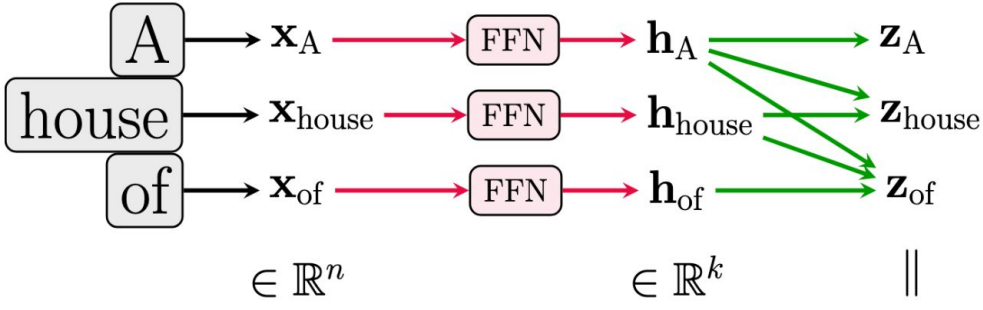
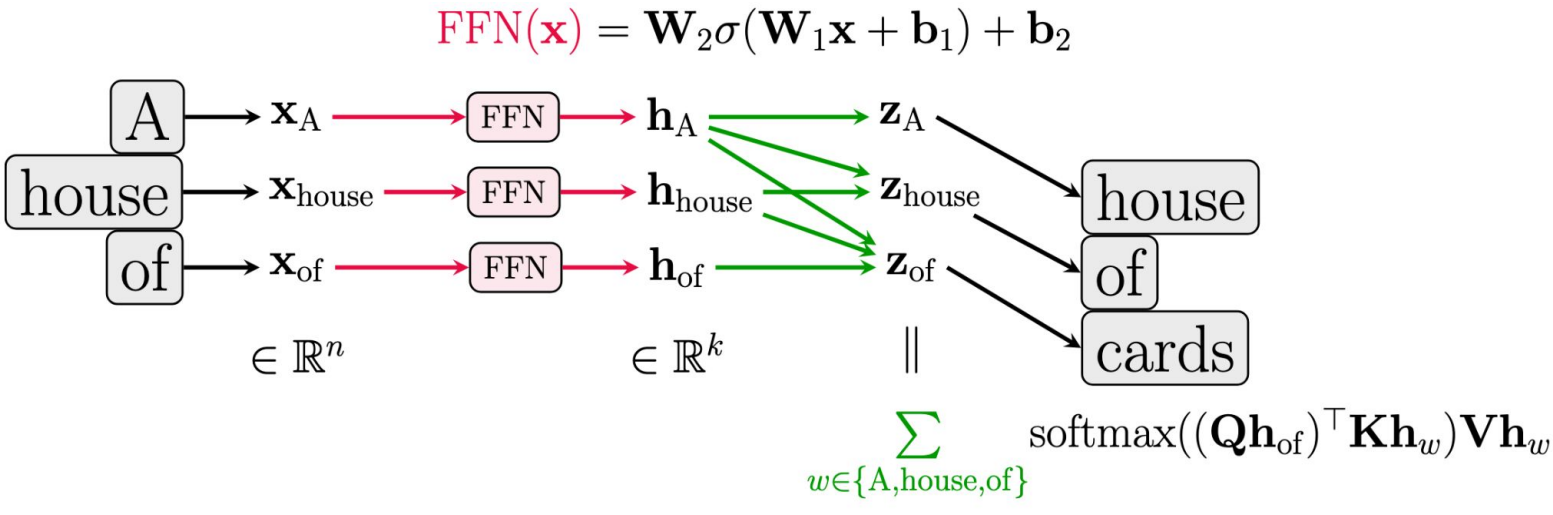


$$\sum_{w \in \{\text{A,house,of}\}} \mathrm{softmax}((\mathbf{Q}\mathbf{h}_{\text{of}})^\top \mathbf{K}\mathbf{h}_w)\mathbf{V}\mathbf{h}_w$$

LLMs as GNNs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A
house
of
$\mathbf{x}_\mathrm{A}$
$\mathbf{x}_\mathrm{house}$
$\mathbf{x}_\mathrm{of}$
FFN
FFN
FFN
$\mathbf{h}_\mathrm{A}$
$\mathbf{h}_\mathrm{house}$
$\mathbf{h}_\mathrm{of}$
$\mathbf{z}_\mathrm{A}$
$\mathbf{z}_\mathrm{house}$
$\mathbf{z}_\mathrm{of}$
house
of
cards
$\in \mathbb{R}^n$
$\in \mathbb{R}^k$
$\|$
$\sum_{w \in \{\mathrm{A,house,of}\}} \mathrm{softmax}((\mathbf{Q}\mathbf{h}_\mathrm{of})^\top \mathbf{K}\mathbf{h}_w)\mathbf{V}\mathbf{h}_w$

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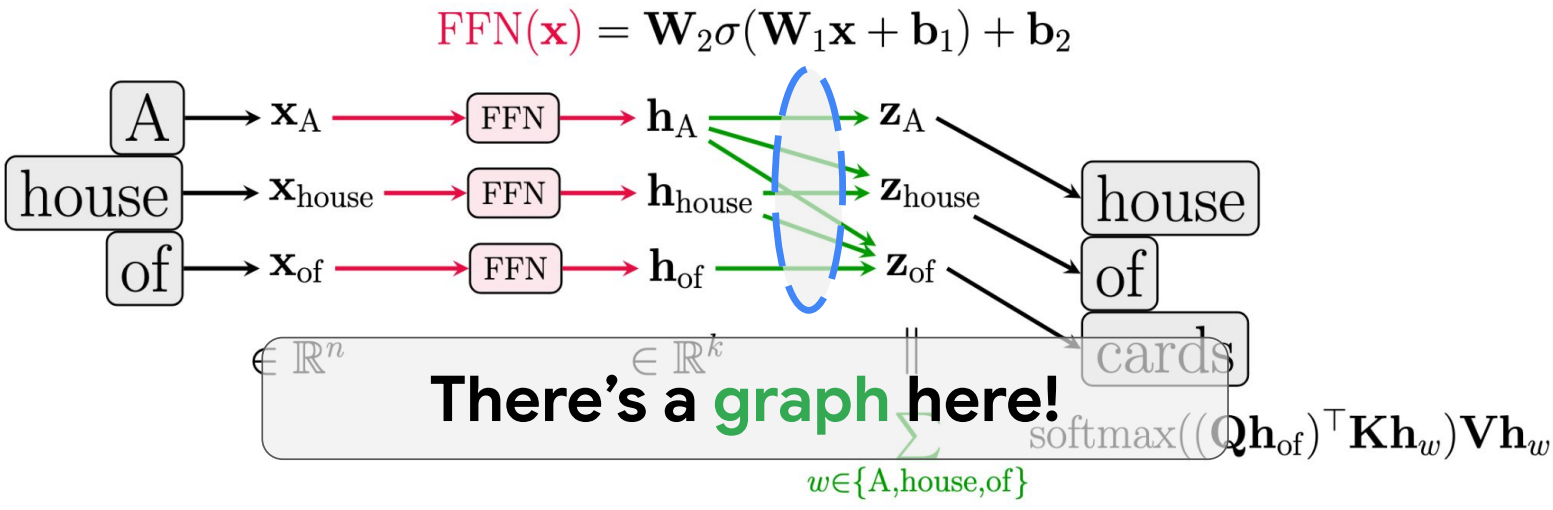
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A
house
of
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FFN
FFN
FFN
$\mathbf{h}_\mathrm{A}$
$\mathbf{h}_\mathrm{house}$
$\mathbf{h}_\mathrm{of}$
$\mathbf{z}_\mathrm{A}$
$\mathbf{z}_\mathrm{house}$
$\mathbf{z}_\mathrm{of}$
house
of
cards
$\in \mathbb{R}^n$
$\in \mathbb{R}^k$
There's a graph here!
$\mathrm{softmax}((\mathbf{Q}\mathbf{h}_\mathrm{of})^\top \mathbf{K}\mathbf{h}_w)\mathbf{V}\mathbf{h}_w$
$\sum$
$w \in \{\mathrm{A}, \mathrm{house}, \mathrm{of}\}$

LLMs as **MPNNs**!

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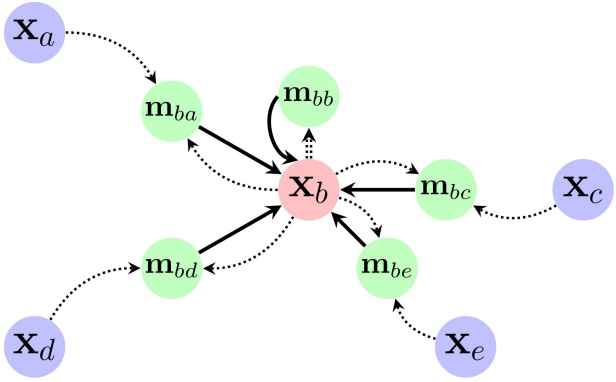

Message-passing

Nodes are tokens, $\mathcal{V} = \{1, \ldots, N\}$

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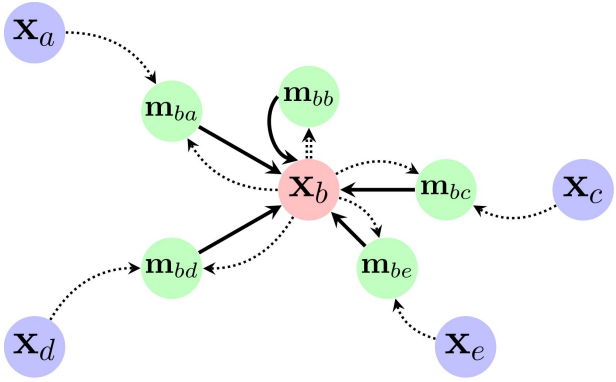

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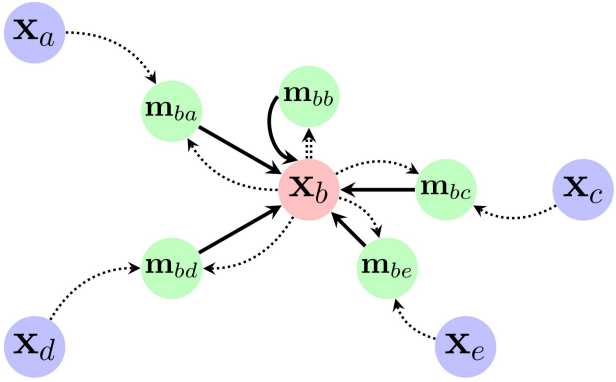

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How about (aggregating) messages?

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Dot-product self-attention as **message passing**

$$\mathbf{z}_u = \sum_{v \in \mathcal{N}_u} \alpha_{uv} \mathbf{v}_v$$

Dot-product self-attention as **message passing**

$$\mathbf{z}_u = \sum_{v \in \mathcal{N}_u} \alpha_{uv} \mathbf{v}_v$$

$$\alpha_{uv} = \frac{\exp e_{uv}}{\sum_{w \in \mathcal{N}_u} \exp e_{uw}}$$

$$\mathbf{v}_v = \mathbf{V}\mathbf{x}_v$$

Dot-product self-attention as **message passing**

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$$\alpha_{uv} = \frac{\exp e_{uv}}{\sum_{w \in \mathcal{N}_u} \exp e_{uw}} \qquad e_{uv} = \mathbf{q}_u^\top \mathbf{k}_v = (\mathbf{Q}\mathbf{x}_u)^\top (\mathbf{K}\mathbf{x}_v) \qquad \mathbf{v}_v = \mathbf{V}\mathbf{x}_v$$

Dot-product self-attention as **message passing**

$$\mathbf{z}_u = \frac{\sum_{v \in \mathcal{N}_u} (\exp e_{uv}) \mathbf{v}_v}{\sum_{v \in \mathcal{N}_u} \exp e_{uv}}$$

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$$\mathbf{z}_u = \frac{\sum_{v \in \mathcal{N}_u} (\exp e_{uv}) \mathbf{v}_v}{\sum_{v \in \mathcal{N}_u} \exp e_{uv}} = \frac{\sum_{v \in \mathcal{N}_u} \mathbf{n}_{uv}}{\sum_{v \in \mathcal{N}_u} d_{uv}}$$

$$\mathbf{m}_{uv} = (\mathbf{n}_{uv}, d_{uv}) = ((\exp \mathbf{q}_u^\top \mathbf{k}_v) \mathbf{v}_v, \exp \mathbf{q}_u^\top \mathbf{k}_v)$$

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$$(\mathbf{n}, d) \oplus (\mathbf{n}', d') = (\mathbf{n} + \mathbf{n}', d + d')$$

Note this does not perform the **division n/d**,
but we can delegate that to ϕ!

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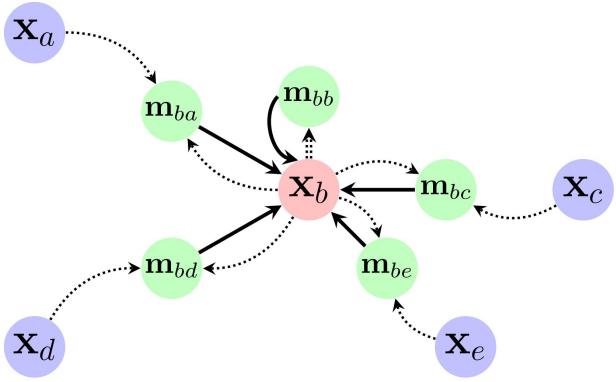

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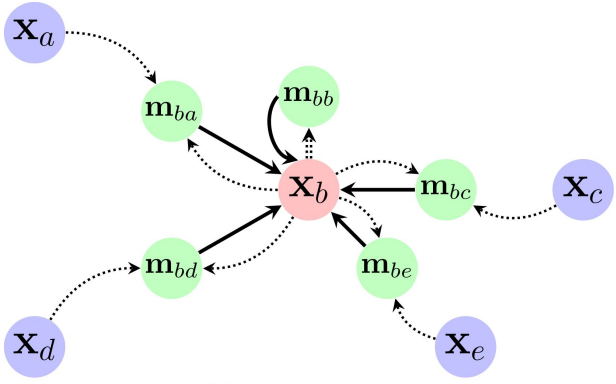

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Message function, ψ, in **two parts**

$$\psi(\mathbf{x}_u, \mathbf{x}_v) = (\exp((\mathbf{Q}\mathbf{x}_u)^\top \mathbf{K}\mathbf{x}_v)\mathbf{V}\mathbf{x}_v, \exp((\mathbf{Q}\mathbf{x}_u)^\top \mathbf{K}\mathbf{x}_v))$$

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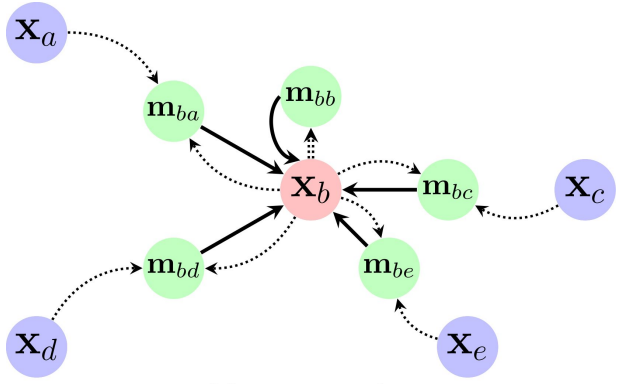

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Aggregation function, $\oplus$, **adds** both

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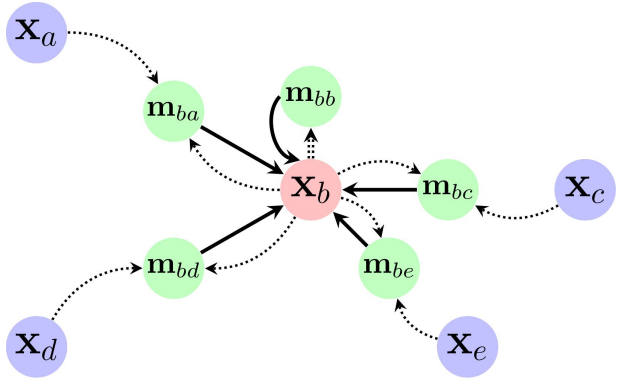

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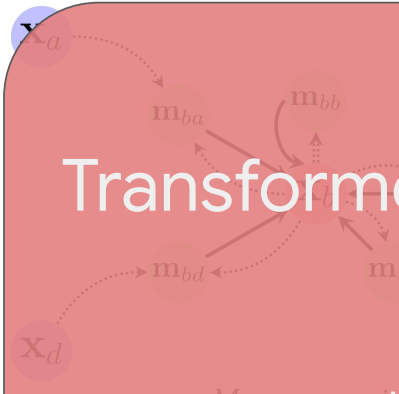

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Transformers **are** Graph Neural Networks

(see **Joshi** (2025);
also **GATs** (Veličković *et al.*, ICLR'18))

Why ***attentional***? Some “conversation starters”:

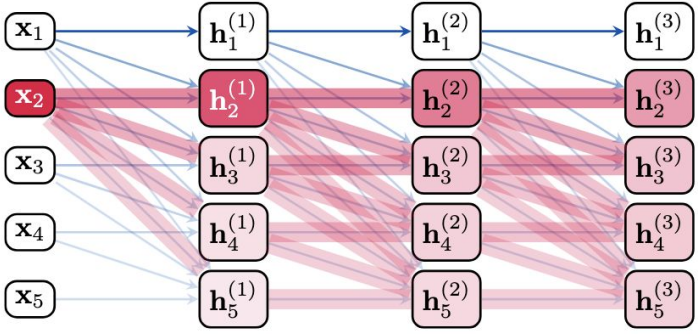
- Tokens are indices into a vocabulary, and **meaningless** without **context**
  → **complex** messages not needed, weighted sums suffice

- Attention requires **no vectors** to be materialised on **edges** — only scalars
  → more easily **scales** to billions of parameters at scale than MPNNs

- Computations (eg. full matrix products) align very well with our **hardware**
  → GNNs that are currently winning the **hardware lottery** (Hooker, CACM’21)

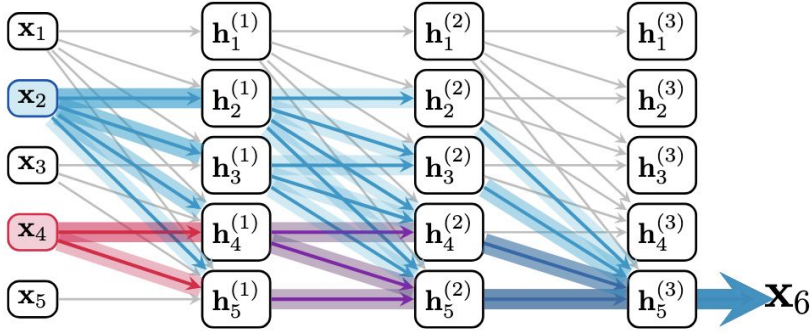
This perspective helps understand common LLM issues!

Overmixing

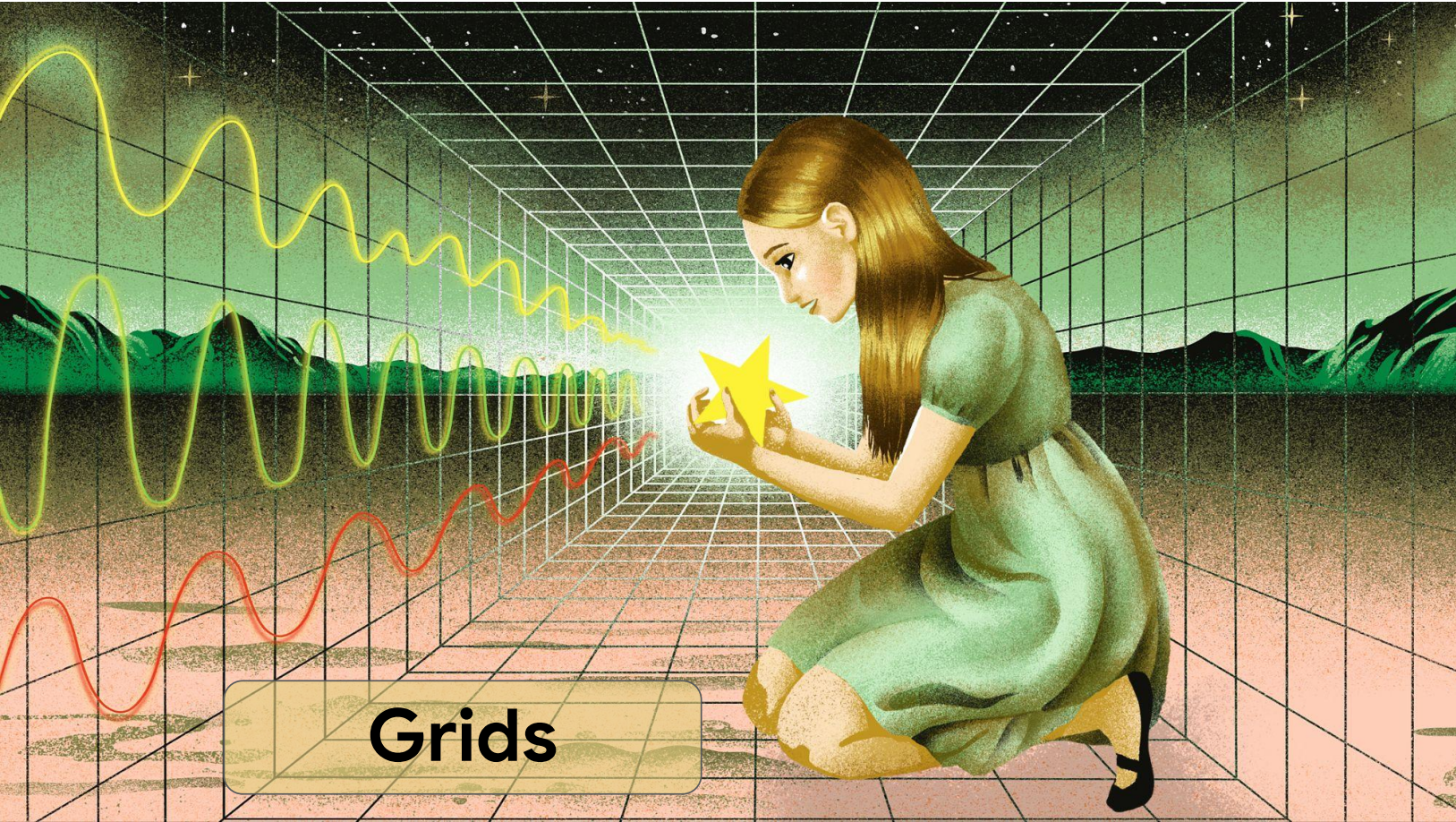


Barbero, Arroyo *et al.*
(COLM'25)

Oversquashing

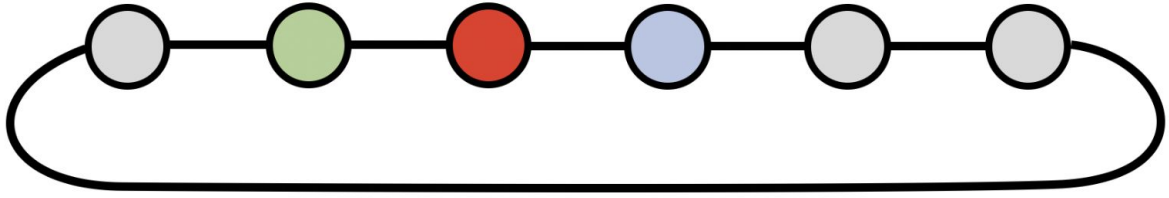


Barbero *et al.*
(NeurIPS'25)


Grids

Like a graph, but more regular ;)

We can look at “special” **permutations** now!
(circular) **Translation equivariance!**



Rigid neighbourhoods: we don’t need local permutation invariance anymore!

$$\phi(\mathbf{x}_{u-1}, \mathbf{x}_u, \mathbf{x}_{u+1}) = \theta_{-1}\mathbf{x}_{u-1} + \theta_0\mathbf{x}_u + \theta_1\mathbf{x}_{u+1}$$

Circulant matrices

Collecting the parameters into a **circulant** matrix:

$$\mathbf{F}(\mathbf{X}) = \begin{bmatrix} \theta_0 & \theta_1 & & & \theta_{-1} \\ \theta_{-1} & \theta_0 & \theta_1 & & \\ & \ddots & \ddots & \ddots & \\ & & \theta_{-1} & \theta_0 & \theta_1 \\ \theta_1 & & & \theta_{-1} & \theta_0 \end{bmatrix} \begin{bmatrix} — & \mathbf{x}_0 & — \\ — & \mathbf{x}_1 & — \\ & \vdots & \\ — & \mathbf{x}_{n-2} & — \\ — & \mathbf{x}_{n-1} & — \end{bmatrix}$$

Circulants are equivariant to circular shifts!

$$\mathbf{S}\mathbf{C}(\boldsymbol{\theta})\mathbf{x} = \mathbf{C}(\boldsymbol{\theta})\mathbf{S}\mathbf{x}$$

Circulant matrices

C

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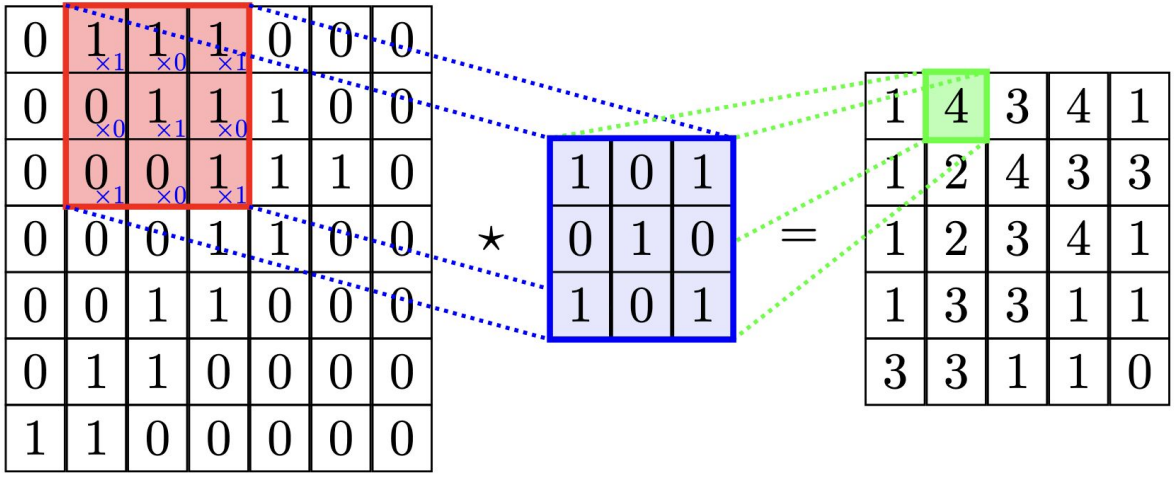
All linear shift equivariant functions **must** be expressible as **circulants**

"All 'true' image models must be CNNs"

Circulants are equivariant to circular shifts!

$$\mathbf{S}\mathbf{C}(\boldsymbol{\theta})\mathbf{x} = \mathbf{C}(\boldsymbol{\theta})\mathbf{S}\mathbf{x}$$

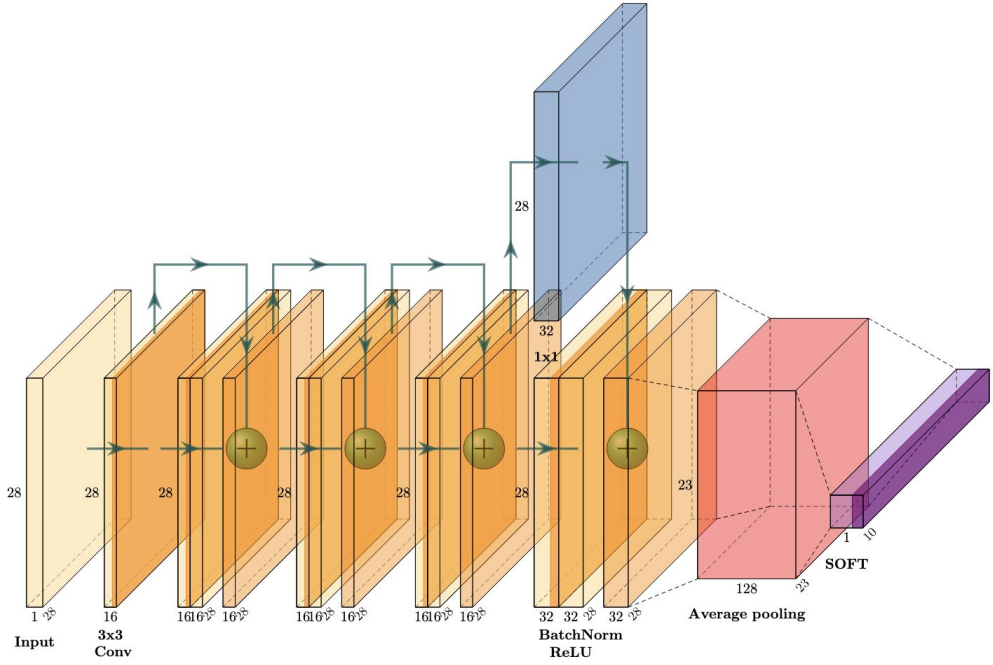
The picture easily generalises to k-D grids

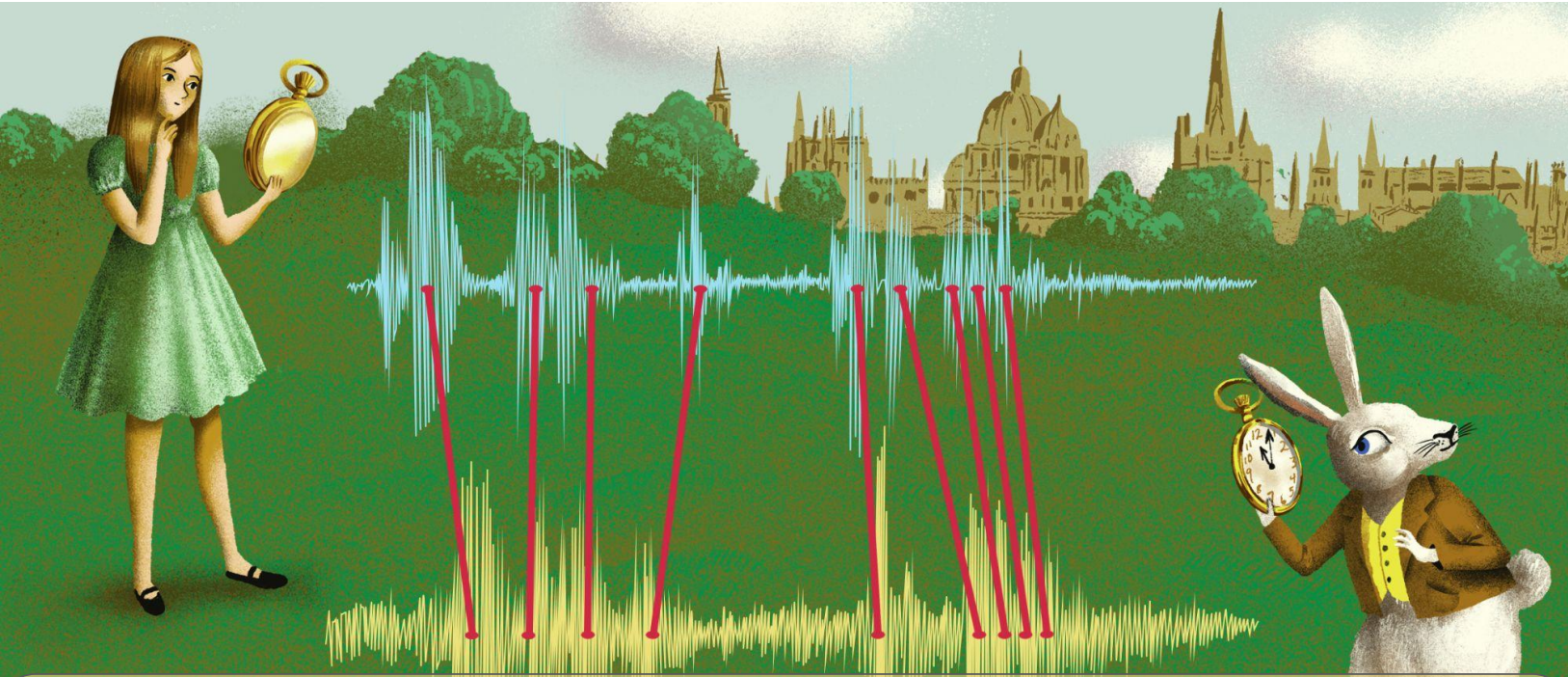


$\mathbf{x}$ $\qquad$ $\mathbf{C}(\boldsymbol{\theta})$ $\qquad$ $\mathbf{x} \star \boldsymbol{\theta}$

$$\overbrace{\boldsymbol{\theta}_{11} + \boldsymbol{\theta}_{13} + \boldsymbol{\theta}_{22} + \boldsymbol{\theta}_{31} + \boldsymbol{\theta}_{33}}$$

...leading to modern CNNs :)





Works for RNNs too:
***time warping* inv. → gating! [GRU/LSTM]**

Groups

Looking at the continuous version of the 1D CNN:

$$(x \star \theta)(u) = \langle x, S_u \theta \rangle = \int_{\mathbb{R}} x(v)\theta(u+v)\mathrm{d}v$$

Could we generalise this for arbitrary (homogeneous) domains and groups?

Looking at the continuous version of the 1D CNN:

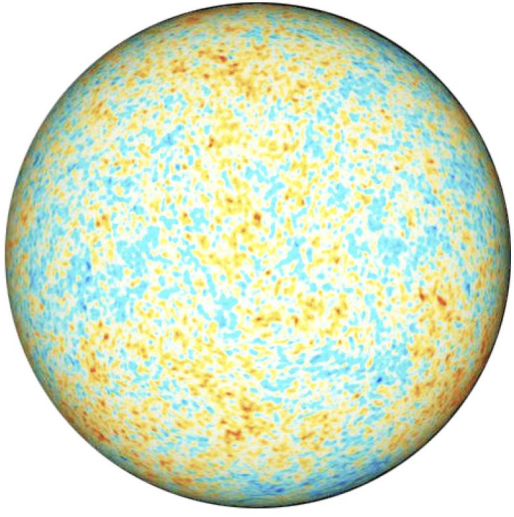
$$(x \star \theta)(u) = \langle x, S_u \theta \rangle = \int_{\mathbb{R}} x(v)\theta(u+v)\mathrm{d}v$$

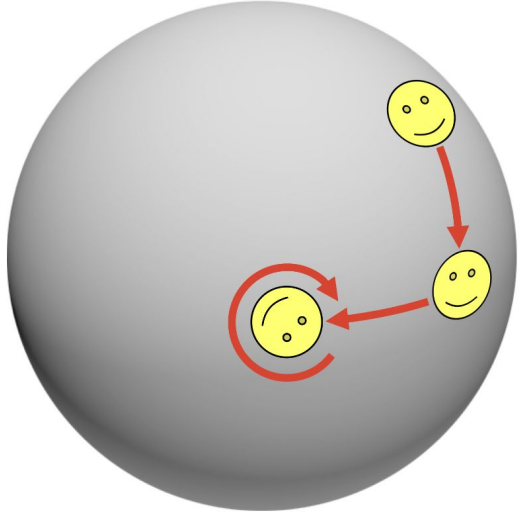
$$(x \star \theta)(g) = \langle x, \rho(g)\theta \rangle = \int_{\Omega} x(u)\theta(g^{-1}u)\mathrm{d}u$$

NB: Signals are now over G, not Ω!!!
It just happened that for shift equivariance, G = Ω = $\mathbb{R}$.

Let's look at a more "exotic" domain 🌐

Sphere, $\Omega = S^2$, with **3D rotation symmetries**, $G = SO(3)$





Spherical CNNs, Part 1: $S^2 \rightarrow SO(3)$

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$$(x \star \theta)(\mathbf{R}) = \int_{\mathbb{S}^2} x(\mathbf{u})\theta(\mathbf{R}^{-1}\mathbf{u})\mathrm{d}\mathbf{u}$$

For rotation matrices **R** and filters $\theta : S^2 \rightarrow \mathbb{R}$

This only covers layer 1 — how can we go deep???

$$(x \star \theta)(\mathbf{R}) = \int_{\mathbb{S}^2} x(\mathbf{u})\theta(\mathbf{R}^{-1}\mathbf{u})\mathrm{d}\mathbf{u}$$

Fortunately, we can make advantage of our knowledge of group actions to make **G-equivariant layers on G itself**!

This only covers layer 1 — how can we go deep???

$$(x \star \theta)(\mathbf{R}) = \int_{\mathbb{S}^2} x(\mathbf{u})\theta(\mathbf{R}^{-1}\mathbf{u})\mathrm{d}\mathbf{u}$$

Take Ω = G, G = G, signals x : G ⟶ $\mathbb{R}$ and group action:

$$(\rho(g)x)(h) = x(g^{-1}h)$$

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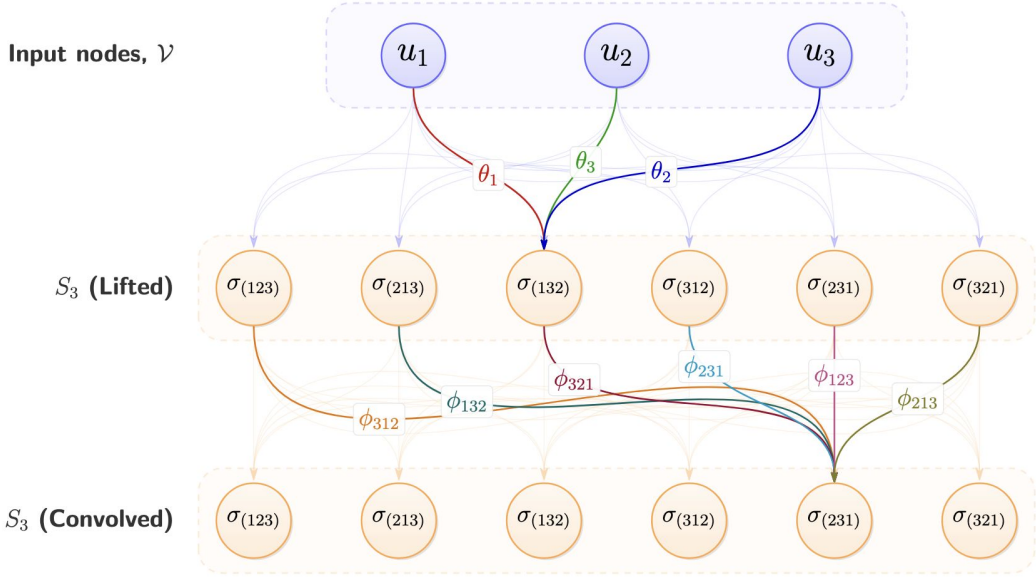
Take Ω = G, G = G, signals x : G ⟶ $\mathbb{R}$ and group action:

$$(\rho(g)x)(h) = x(g^{-1}h)$$

Deep spherical CNNs:

$$((x \star \theta) \star \phi)(\mathbf{R}) = \int_{\mathrm{SO}(3)} (x \star \theta)(\mathbf{Q})\phi(\mathbf{R}^{-1}\mathbf{Q})\mathrm{d}\mathbf{Q}$$

Can we apply this idea to graphs too?



Yes, but... it quickly goes out of hand :)

What would a “group convolutional” GNN look like?

Layer 1 : $V \longrightarrow S_n$

$$(x \star \theta)(\sigma) = \sum_{u \in \mathcal{V}} x(u)\theta(\sigma^{-1}(u))$$

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Layer 2 : $S_n \longrightarrow S_n$

$$((x \star \theta) \star \phi)(\sigma) = \sum_{\sigma' \in S_n} (x \star \theta)(\sigma')\phi(\sigma^{-1} \circ \sigma')$$

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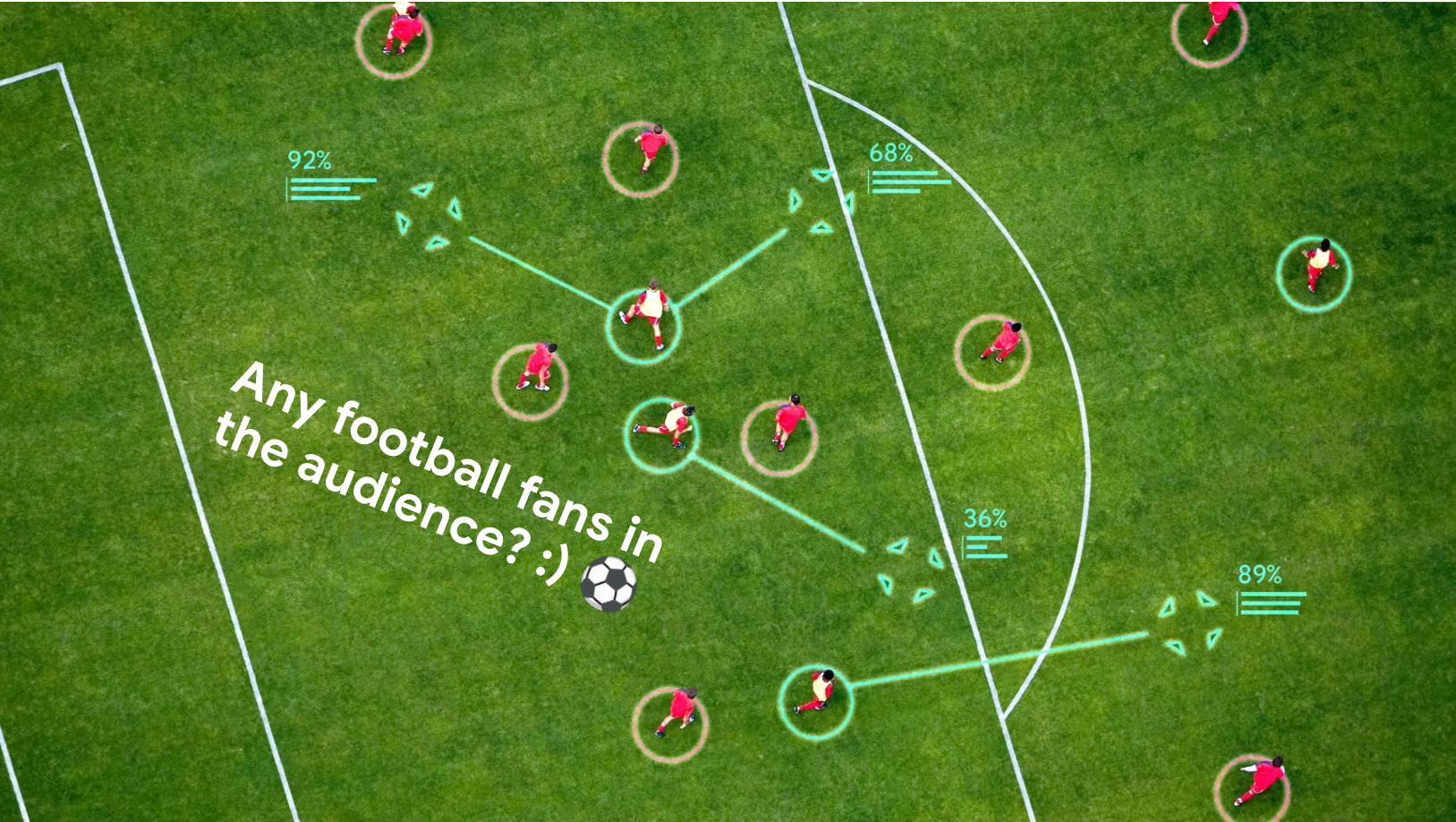
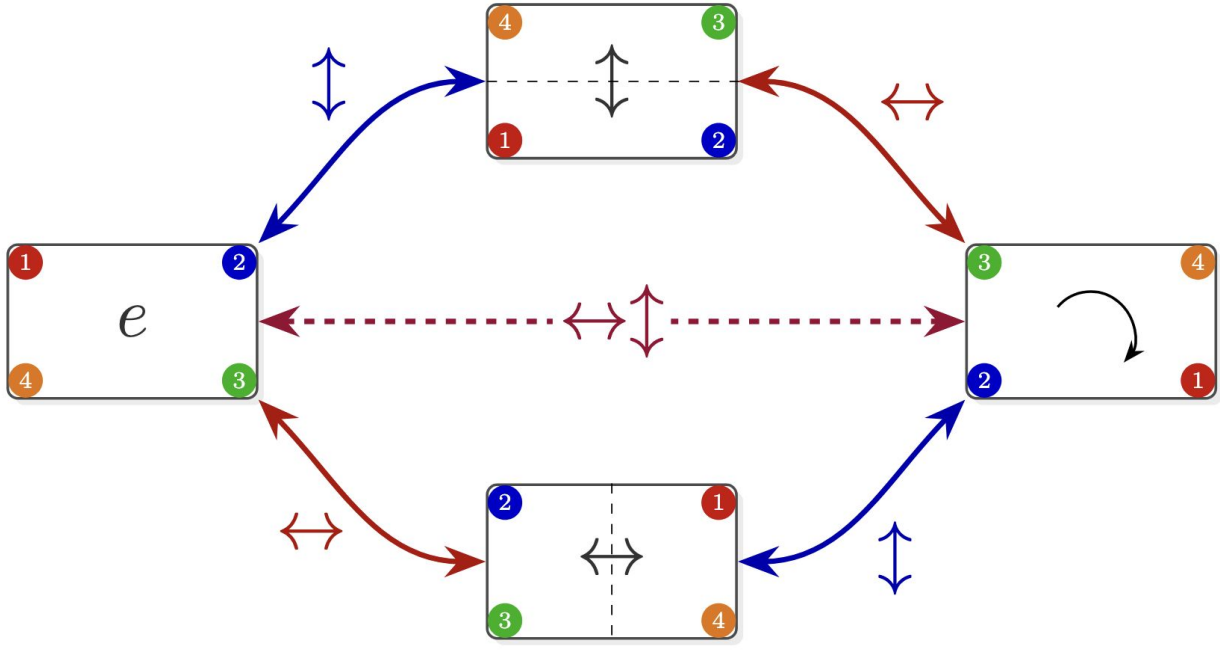
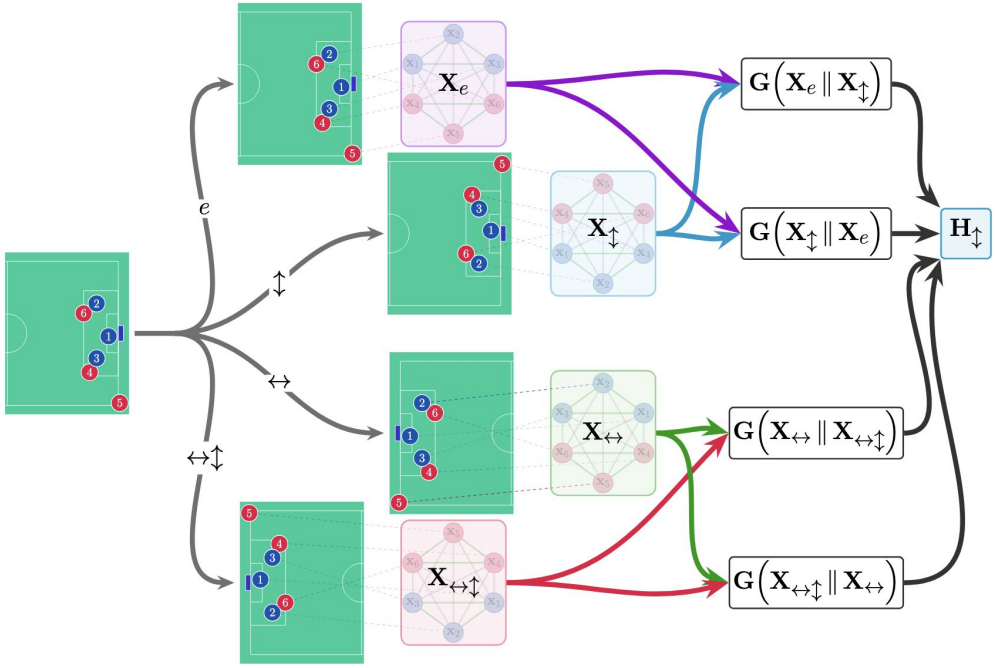
This rapidly gets unwieldy as $n \geq 8$... but hints at a **great wealth** of GNNs not easily expressed as 1-hop message passing!


92%
68%
Any football fans in the audience? :)
36%
89%

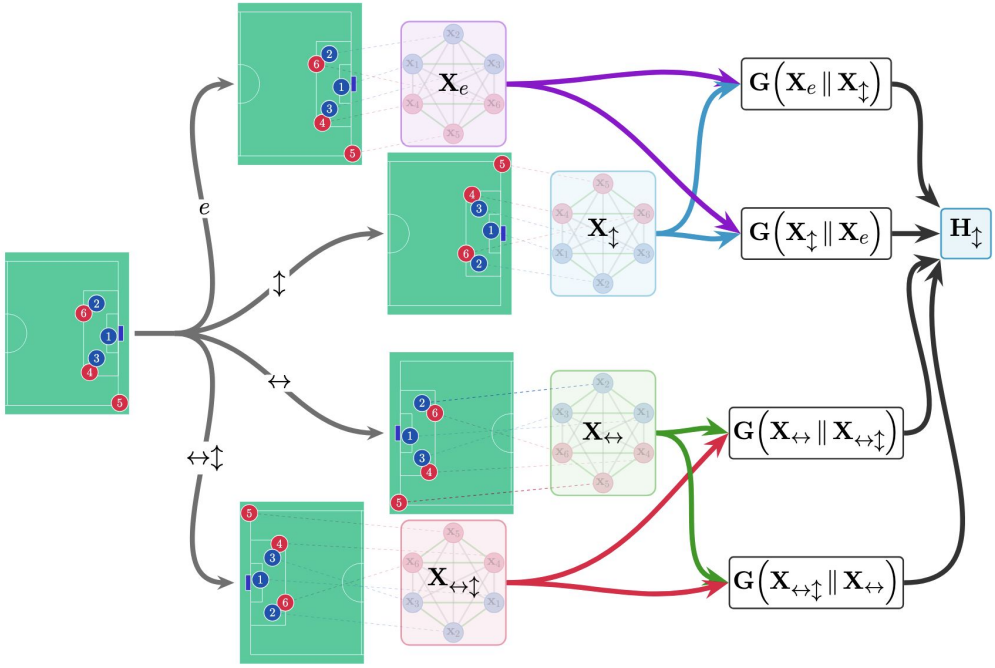
Dihedral group D_2: An (approximate) pitch symmetry

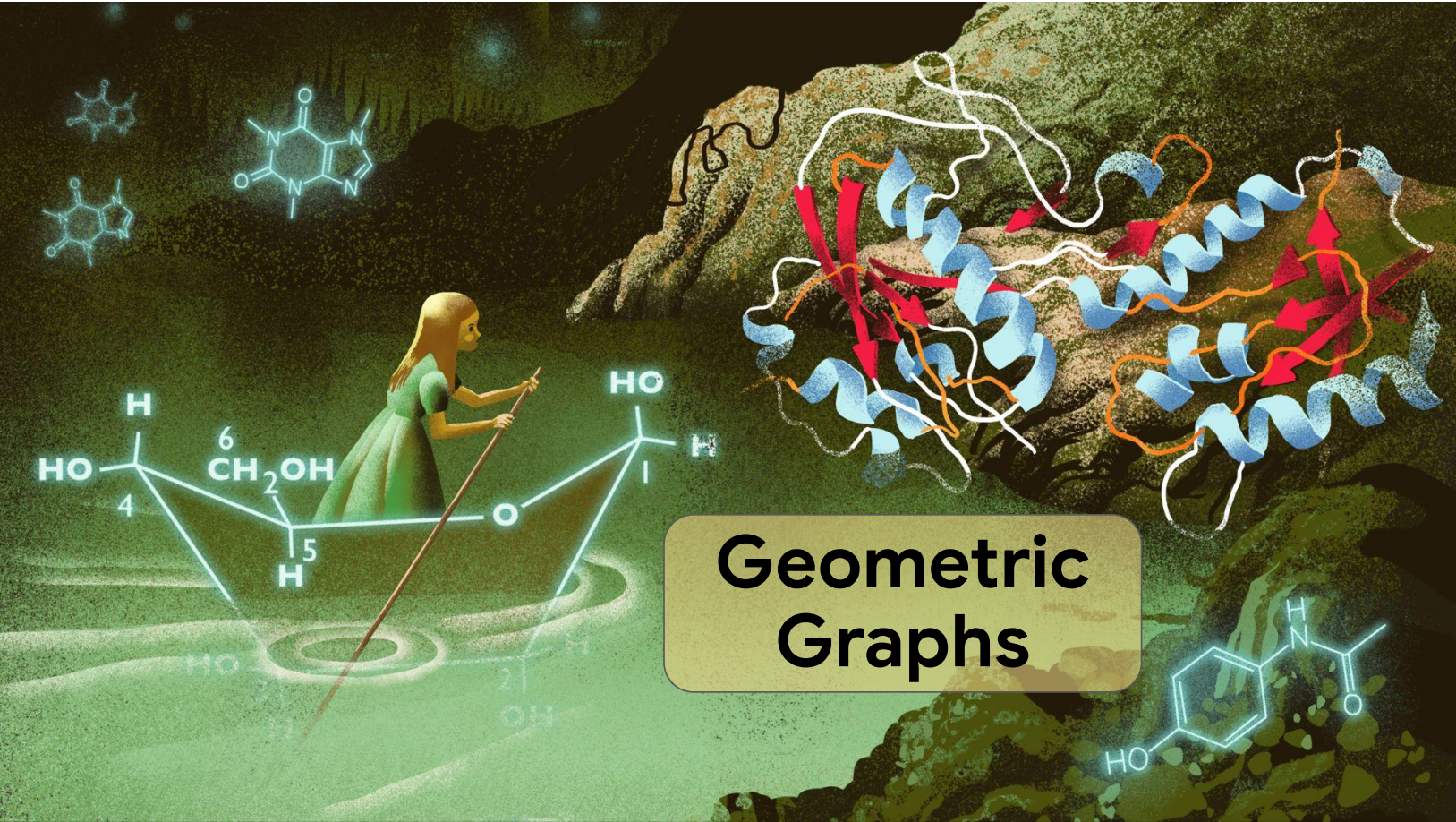


D_2-equivariant group convolution: **TacticAI**

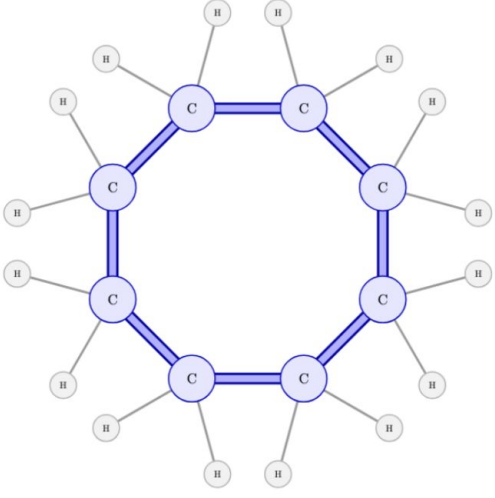


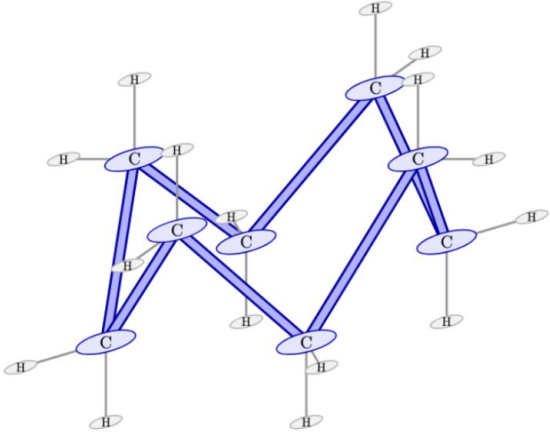
D_2-equivariant group convolution: **TacticAI**

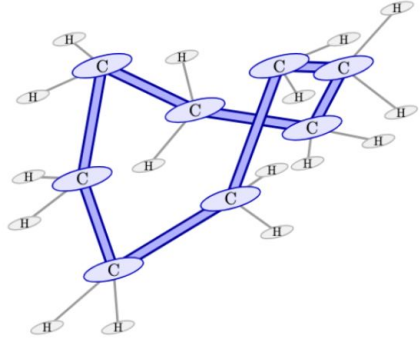



Geometric
Graphs

Thus far we assumed our domain to “float in vacuum” —
but it’s often very much not doing that!







Roto-translation (+reflection) equivariance (E(3)) **with** permutation equivariance —> “Geometric Graphs”

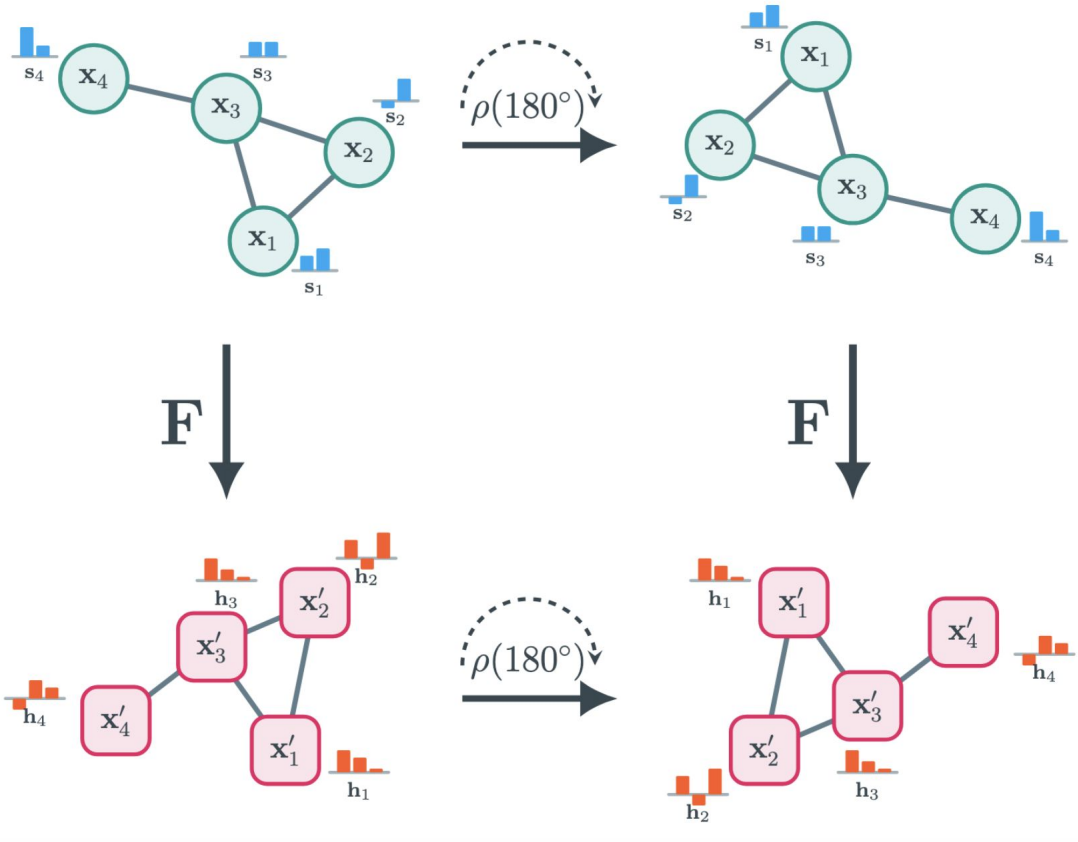
$$\mathbf{F}(\mathbf{PS}, \mathbf{PAP}^\top, \mathbf{PXQ} + \mathbf{t}) = \mathbf{PF}(\mathbf{S}, \mathbf{A}, \mathbf{X})$$

Explicitly splitting

(scalar) node **features**, $\mathbf{s}_u \in \mathbb{R}^k$
from node **coordinates**, $\mathbf{x}_u \in \mathbb{R}^3$

For all:

orthogonal 3D matrices, $\mathbf{Q} \in \mathbb{R}^{3 \times 3}$
& translations, $\mathbf{t} \in \mathbb{R}^3$


$\mathbf{x}_4$
$\mathbf{x}_3$
$\mathbf{x}_2$
$\mathbf{x}_1$
$\mathbf{s}_4$
$\mathbf{s}_3$
$\mathbf{s}_2$
$\mathbf{s}_1$
$\rho(180°)$
$\mathbf{x}_1$
$\mathbf{x}_2$
$\mathbf{x}_3$
$\mathbf{x}_4$
$\mathbf{s}_1$
$\mathbf{s}_2$
$\mathbf{s}_3$
$\mathbf{s}_4$
F
F
$\mathbf{x}'_2$
$\mathbf{x}'_3$
$\mathbf{x}'_4$
$\mathbf{x}'_1$
$\mathbf{h}_3$
$\mathbf{h}_2$
$\mathbf{h}_4$
$\mathbf{h}_1$
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$\mathbf{x}'_1$
$\mathbf{x}'_4$
$\mathbf{x}'_3$
$\mathbf{x}'_2$
$\mathbf{h}_1$
$\mathbf{h}_4$
$\mathbf{h}_3$
$\mathbf{h}_2$

If we only message pass based on **invariant** properties, then we are guaranteed good E(3) behaviours!

$$\mathbf{h}_u = \phi\left(\mathbf{s}_u, \bigoplus_{v \in \mathcal{N}_u} \psi\left(\mathbf{s}_u, \mathbf{s}_v, \|\mathbf{x}_u - \mathbf{x}_v\|\right)\right)$$

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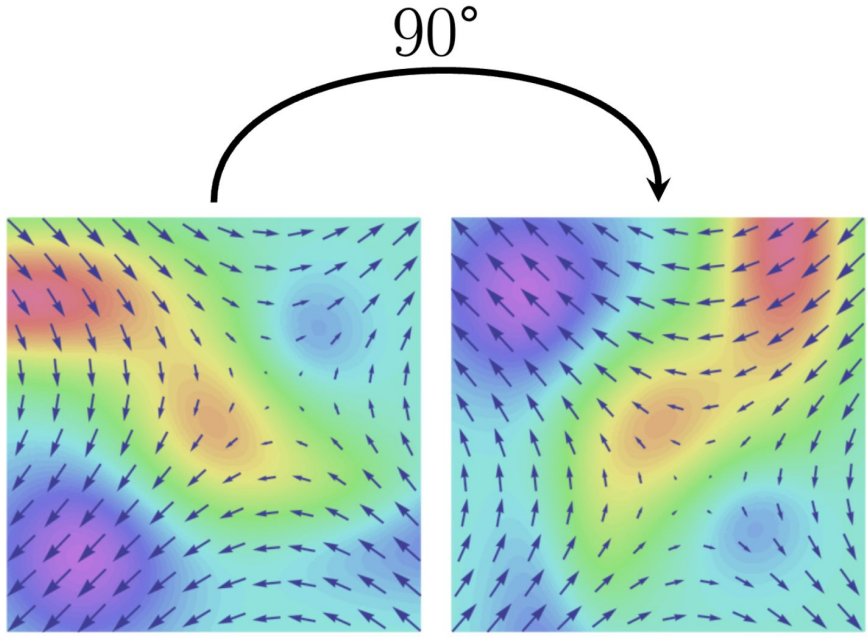
We can also (directionally) update **coordinates**, but we need to multiply them by invariant quantities :)

$$\mathbf{x}'_u = \mathbf{x}_u + \sum_{v \neq u} \beta\left(\mathbf{s}_u, \mathbf{s}_v, \|\mathbf{x}_u - \mathbf{x}_v\|\right)(\mathbf{x}_u - \mathbf{x}_v)$$

What if we have **vector**-structured features?



What if we have **vector**-structured features?

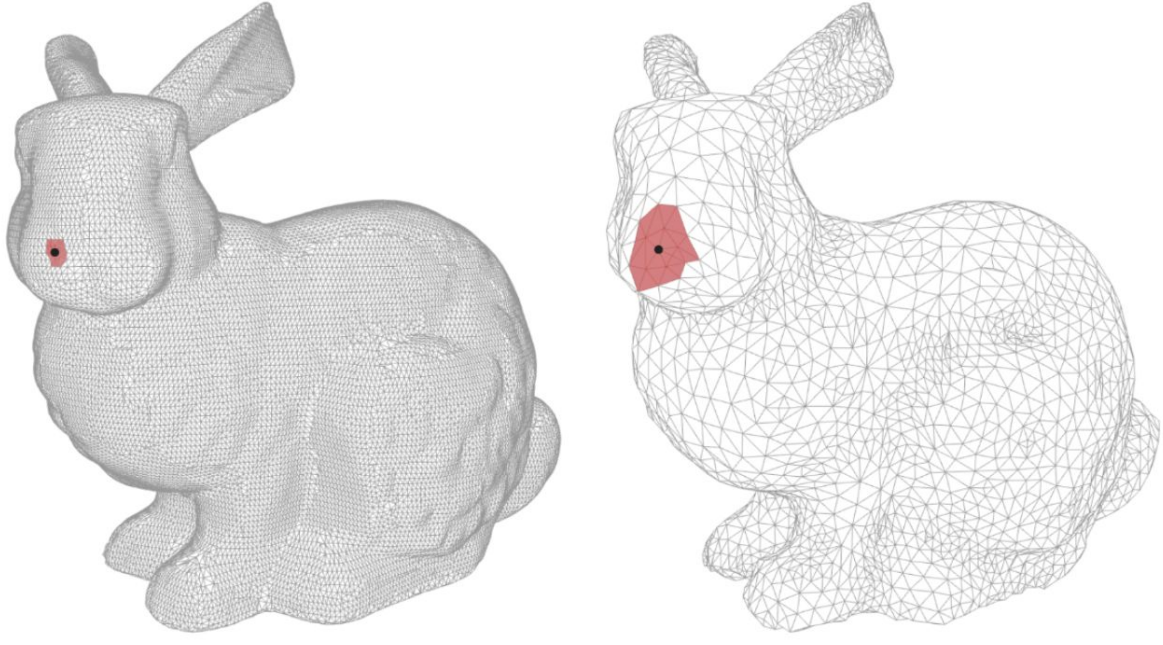
$$\mathbf{F}(\mathbf{PS}, \mathbf{PVQ}, \mathbf{PAP}^\top, \mathbf{PXQ} + \mathbf{t}) = (\mathbf{PH}, \mathbf{PV}'\mathbf{Q}, \mathbf{PX}'\mathbf{Q} + \mathbf{t})$$

Similar reasoning to coordinate updates:

$$\mathbf{v}'_u = \mathbf{v}_u + \sum_{v \neq u} \alpha\left(\mathbf{s}_u, \mathbf{s}_v, \|\mathbf{x}_u - \mathbf{x}_v\|\right) \mathbf{v}_v + \sum_{v \neq u} \beta\left(\mathbf{s}_u, \mathbf{s}_v, \|\mathbf{x}_u - \mathbf{x}_v\|\right) (\mathbf{x}_u - \mathbf{x}_v)$$

Manifolds

What if our geometric graphs are not just a point cloud?



How can we manipulate data on manifolds?

A 2D manifold, Ω, is “locally” like $\mathbb{R}^2$;
we can exploit that to define **filters**!

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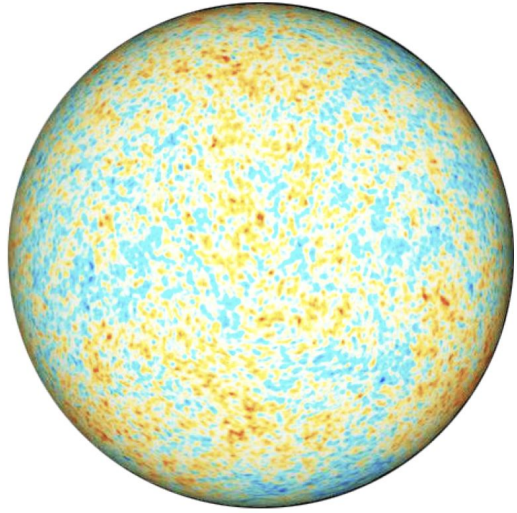
To illustrate key concepts we will be using our **spherical** friend, $\Omega = S^2$



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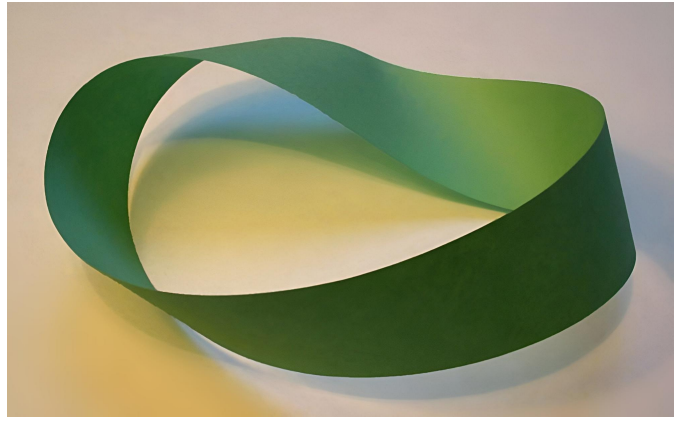
But please keep in mind spheres are *homogeneous* (“every point looks the same”); real manifolds get significantly more **weird** ;)



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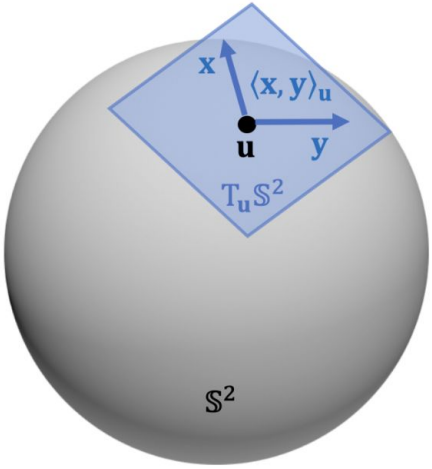
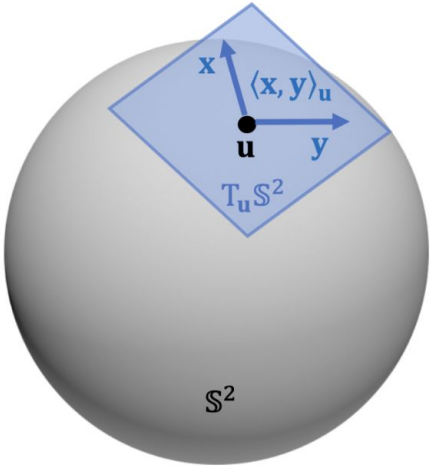
Key terms



Tangent space, $T_u\Omega$

"Local 2D space" around a point, wherein we can manipulate vectors

Key terms



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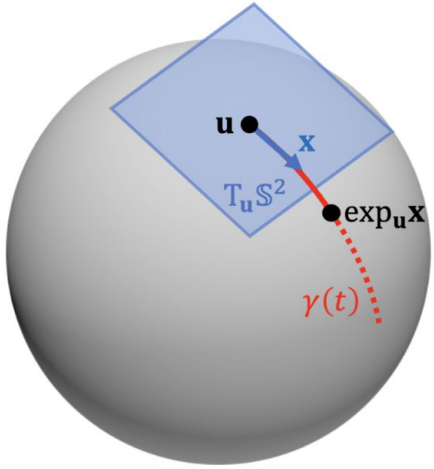
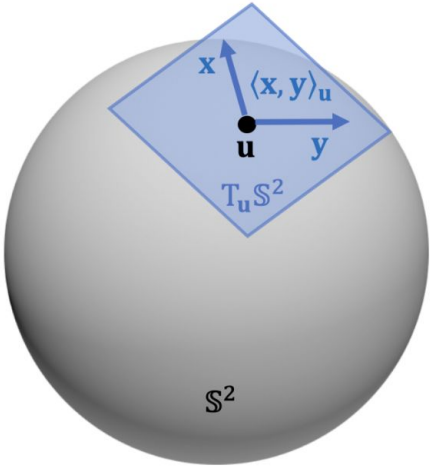


Exponential map, $\exp_u X$

Projecting a tangent vector $X \in T_u\Omega$ back onto Ω, along the **geodesic, γ**

Key terms



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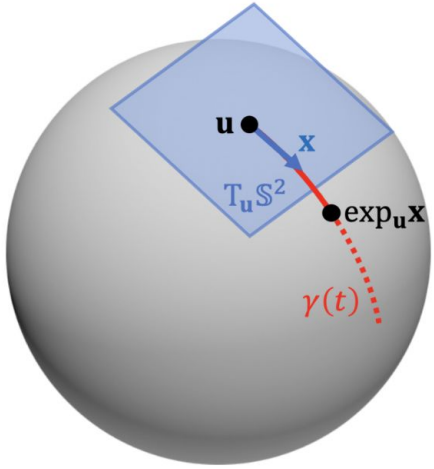

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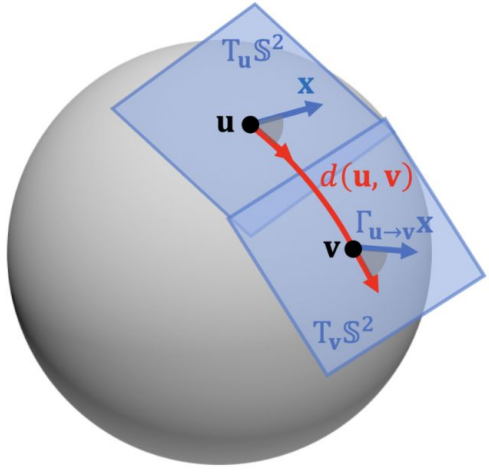

Parallel Transport, $\Gamma_{u \to v} X$

Transporting a tangent vector $X \in T_u\Omega$ along the geodesic, into $\boldsymbol{\Gamma}_{u \to v} X \in T_v\Omega$

We can try defining a convolution in Tangent space:

$$(x \star \theta)(u) = \int_{T_u\Omega} x(\exp_u Y)\theta_u(Y)\, \mathrm{d}Y$$

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But tangent vectors Y are *meaningless* without a “baseline direction” (a **gauge**, $\omega_u : \mathbb{R}^2 \longrightarrow T_u\Omega$) to assign coordinates from!

$$(x \star \theta)(u) = \int_{[0,1]^s} x(\exp_u(\omega_u \mathbf{y}))\theta(\mathbf{y})\, \mathbf{dy}$$

Problem: choice of gauge is totally **arbitrary**

Even on spheres, we cannot get “canonical” set of gauges, due to the Poincaré-Hopf Theorem (“hairy ball theorem”)

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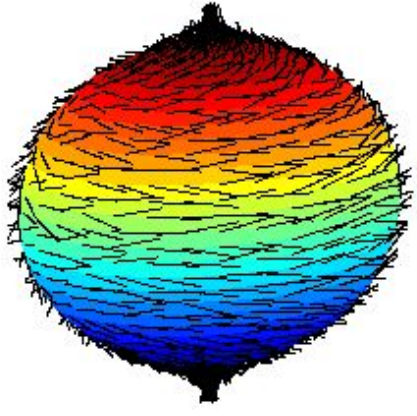

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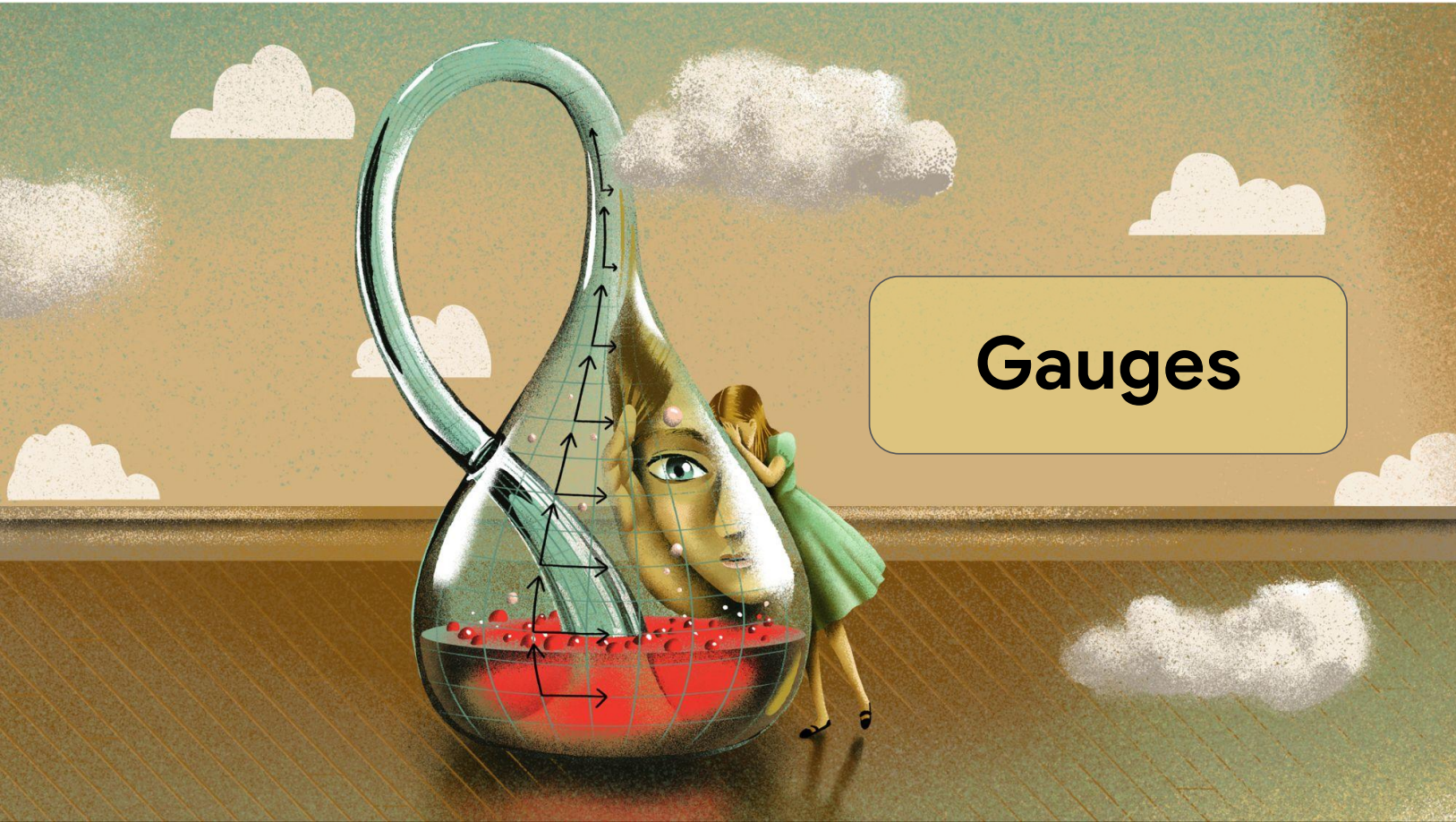
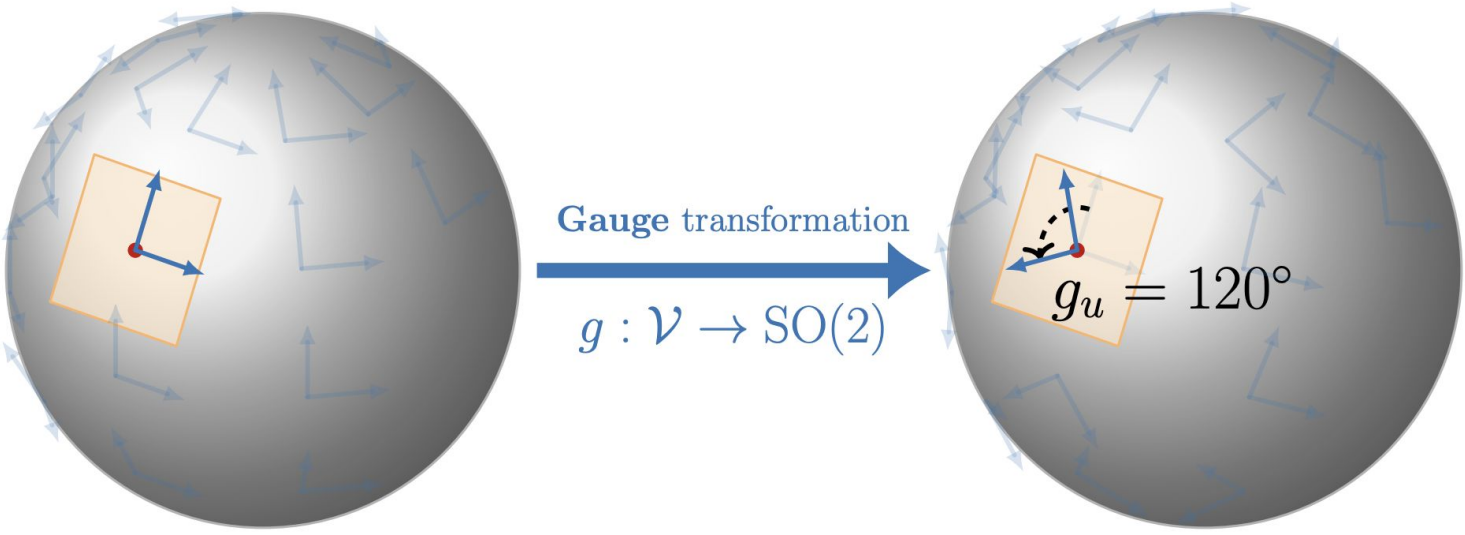
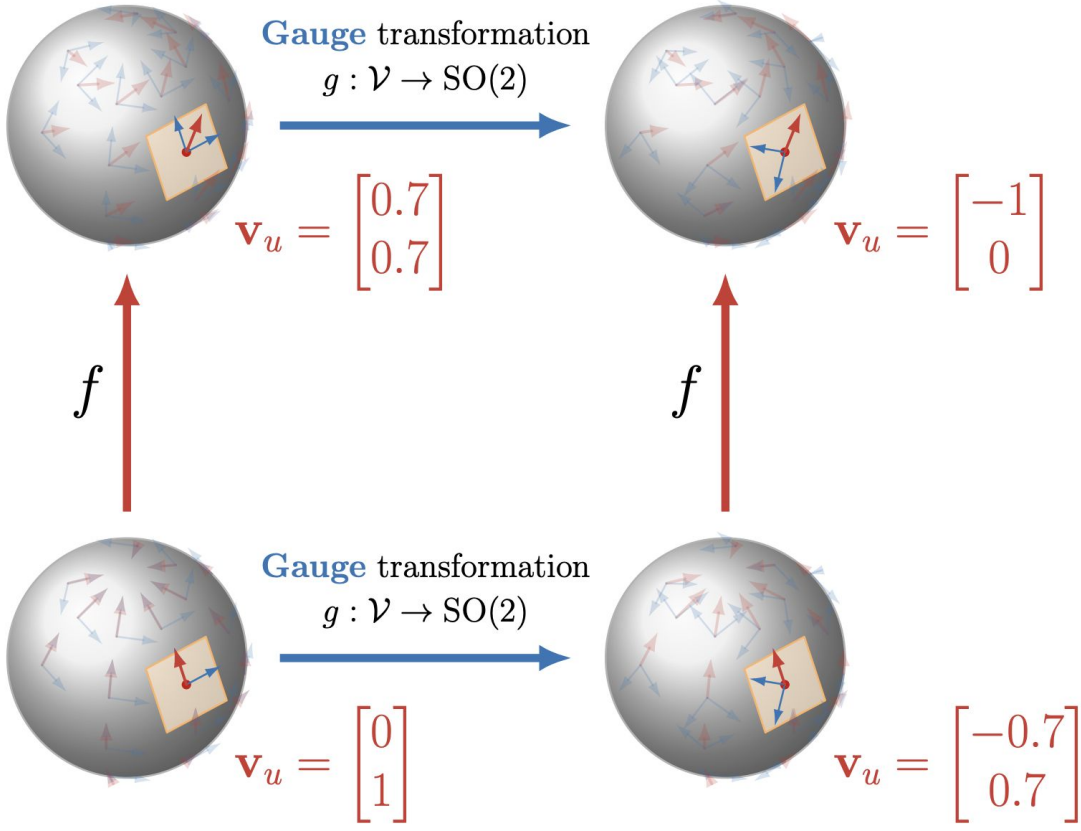
Hence, we need to design models that are *equivariant* to the choice of gauge!


Gauges

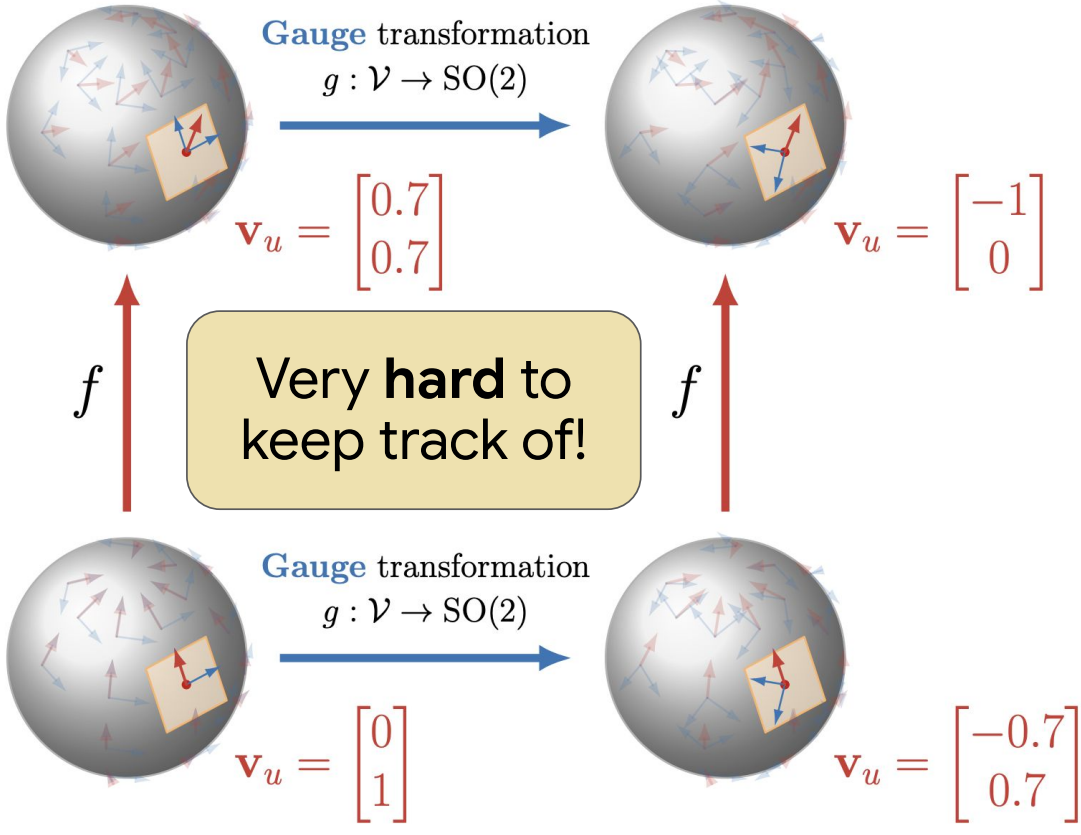
Once again: choice of gauge is totally **arbitrary**



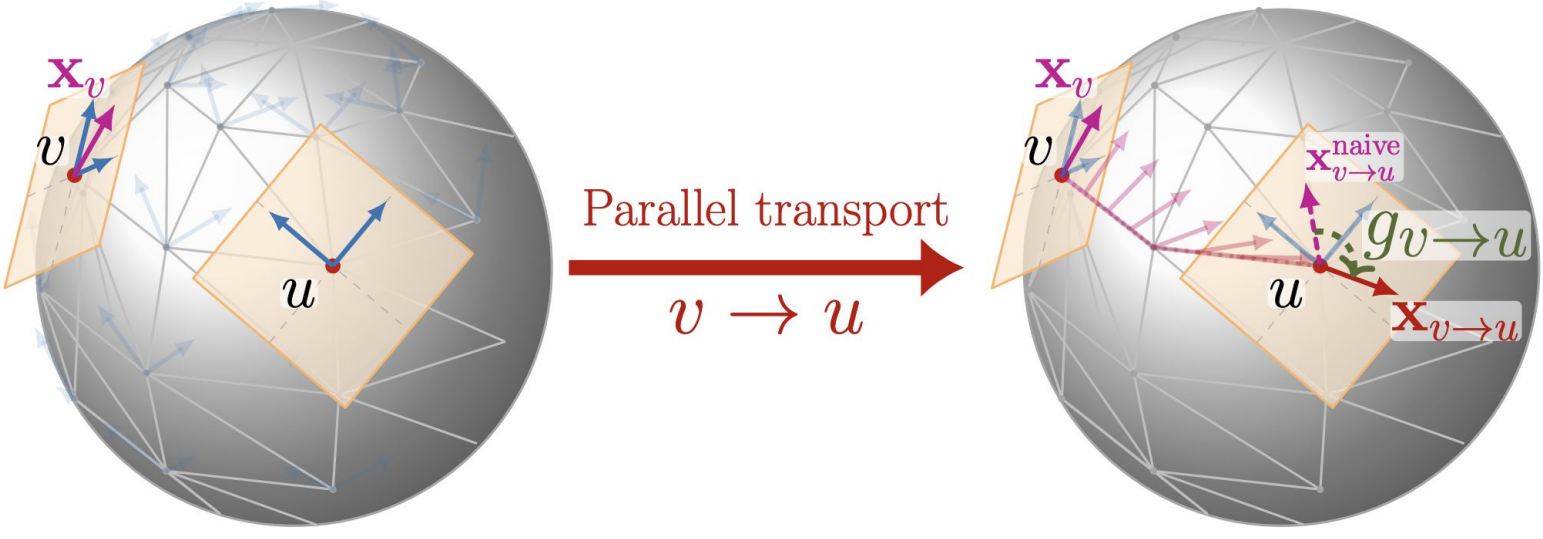
And each gauge choice changes **vector input features**!!!
Our model should be ***equivariant*** to this choice


Gauge transformation
$g : \mathcal{V} \to \mathrm{SO}(2)$
$\mathbf{v}_u = \begin{bmatrix} 0.7 \\ 0.7 \end{bmatrix}$
$\mathbf{v}_u = \begin{bmatrix} -1 \\ 0 \end{bmatrix}$
f
f
Gauge transformation
$g : \mathcal{V} \to \mathrm{SO}(2)$
$\mathbf{v}_u = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$
$\mathbf{v}_u = \begin{bmatrix} -0.7 \\ 0.7 \end{bmatrix}$

Condition across **pairs of gauges** that can transform **arbitrarily**!



Simplify by **parallel transporting** the features first

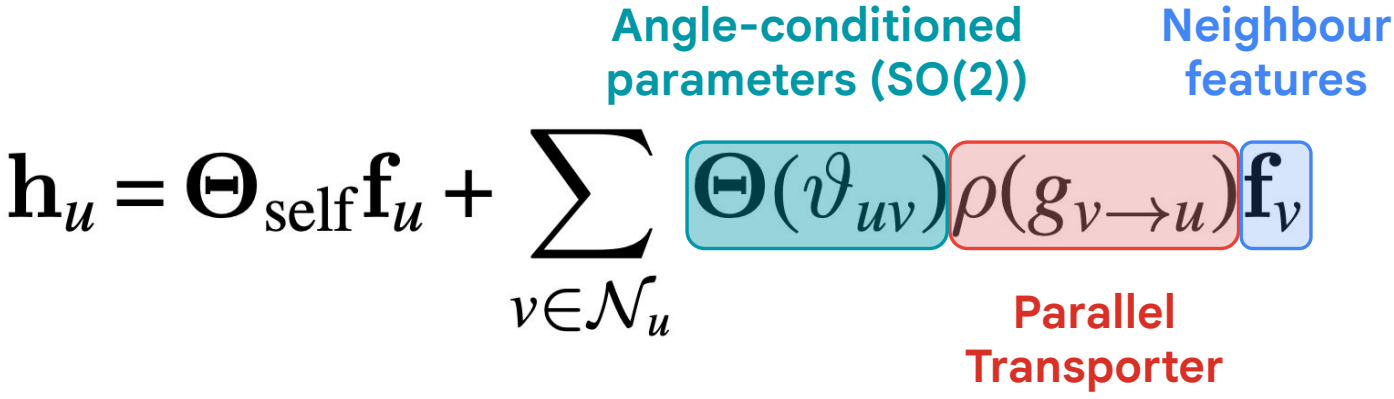


Now, gauge equivariance simply becomes **SO(2) equivariance** to **local** choice of gauge in each receiver!

What we get: GNN with extra pre-computations :)

$$\mathbf{h}_u = \mathbf{\Theta}_{\text{self}} \mathbf{f}_u + \sum_{v \in \mathcal{N}_u} \mathbf{\Theta}(\vartheta_{uv}) \rho(g_{v \to u}) \mathbf{f}_v$$

What we get: GNN with extra pre-computations :)



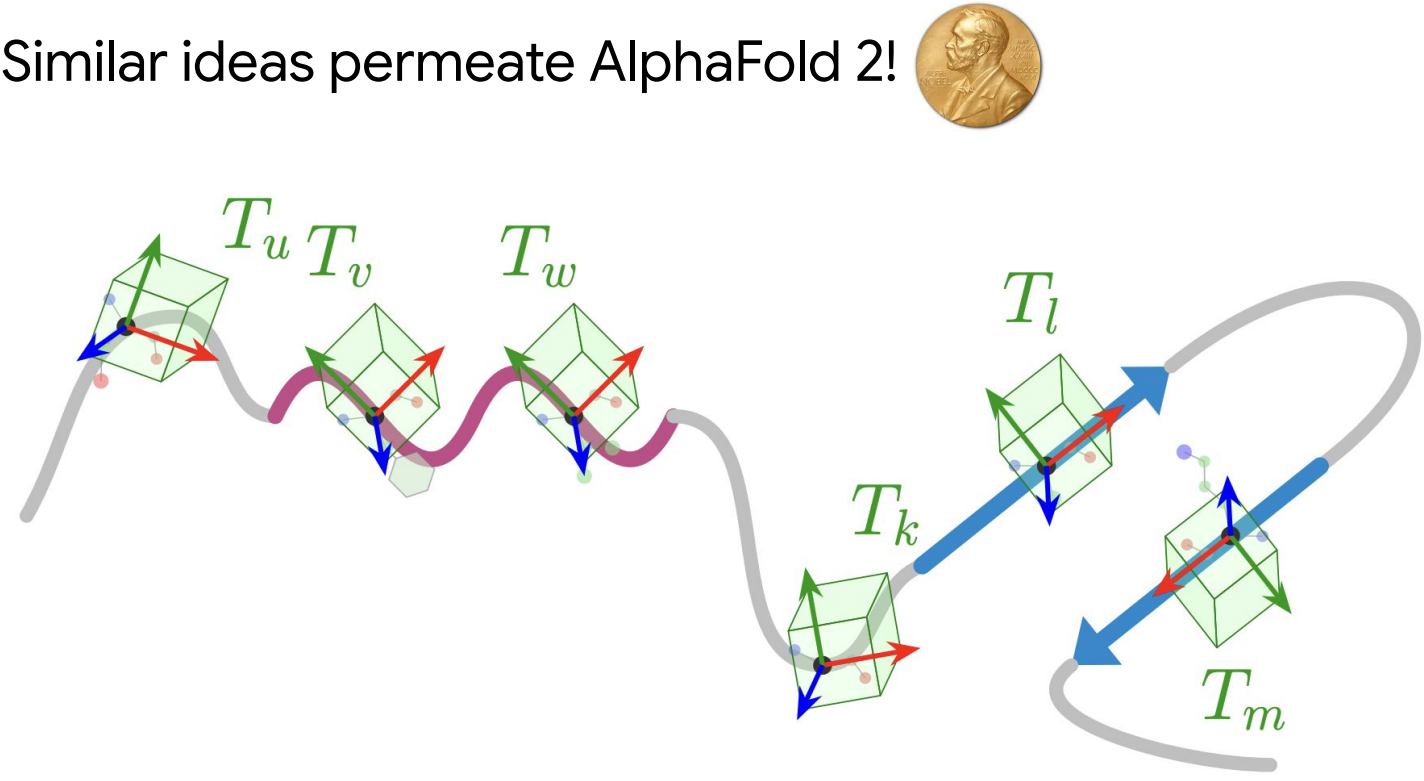
What we get: GNN with extra pre-computations :)

"It's just a graph with extra decorations"

$$\mathbf{h}_u = \Theta_{\text{self}}\mathbf{f}_u + \sum_{v \in \mathcal{N}_u} \Theta(\vartheta_{vw})\rho(g_{v\to u})\mathbf{f}_v$$

No matter how exotic the domain gets,
once data lands on the computer,
we need to discretise it, obtaining **nodes** :)

Similar ideas permeate AlphaFold 2!



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Thank you! Hvala! Mulțumesc!

Now it's your turn to study geometry :)

Petar Veličković
petarv@google.com
https://petar-v.com

Google DeepMind